

and *still* gives all the antiderivatives there are. For this reason, we recommend that you go right to the final form even if you elect to integrate term-by-term. Write

$$\begin{aligned}\int (x^2 - 2x + 5) dx &= \int x^2 dx - \int 2x dx + \int 5 dx \\ &= \frac{x^3}{3} - x^2 + 5x + C.\end{aligned}$$

Find the simplest antiderivative you can for each part and add the arbitrary constant of integration at the end. ■

EXERCISES 4.8

Finding Antiderivatives

In Exercises 1–16, find an antiderivative for each function. Do as many as you can mentally. Check your answers by differentiation.

- | | | |
|--------------------------------|---|---|
| 1. a. $2x$ | b. x^2 | c. $x^2 - 2x + 1$ |
| 2. a. $6x$ | b. x^7 | c. $x^7 - 6x + 8$ |
| 3. a. $-3x^{-4}$ | b. x^{-4} | c. $x^{-4} + 2x + 3$ |
| 4. a. $2x^{-3}$ | b. $\frac{x^{-3}}{2} + x^2$ | c. $-x^{-3} + x - 1$ |
| 5. a. $\frac{1}{x^2}$ | b. $\frac{5}{x^2}$ | c. $2 - \frac{5}{x^2}$ |
| 6. a. $-\frac{2}{x^3}$ | b. $\frac{1}{2x^3}$ | c. $x^3 - \frac{1}{x^3}$ |
| 7. a. $\frac{3}{2}\sqrt{x}$ | b. $\frac{1}{2\sqrt{x}}$ | c. $\sqrt{x} + \frac{1}{\sqrt{x}}$ |
| 8. a. $\frac{4}{3}\sqrt[3]{x}$ | b. $\frac{1}{3\sqrt[3]{x}}$ | c. $\sqrt[3]{x} + \frac{1}{\sqrt[3]{x}}$ |
| 9. a. $\frac{2}{3}x^{-1/3}$ | b. $\frac{1}{3}x^{-2/3}$ | c. $-\frac{1}{3}x^{-4/3}$ |
| 10. a. $\frac{1}{2}x^{-1/2}$ | b. $-\frac{1}{2}x^{-3/2}$ | c. $-\frac{3}{2}x^{-5/2}$ |
| 11. a. $-\pi \sin \pi x$ | b. $3 \sin x$ | c. $\sin \pi x - 3 \sin 3x$ |
| 12. a. $\pi \cos \pi x$ | b. $\frac{\pi}{2} \cos \frac{\pi x}{2}$ | c. $\cos \frac{\pi x}{2} + \pi \cos x$ |
| 13. a. $\sec^2 x$ | b. $\frac{2}{3} \sec^2 \frac{x}{3}$ | c. $-\sec^2 \frac{3x}{2}$ |
| 14. a. $\csc^2 x$ | b. $-\frac{3}{2} \csc^2 \frac{3x}{2}$ | c. $1 - 8 \csc^2 2x$ |
| 15. a. $\csc x \cot x$ | b. $-\csc 5x \cot 5x$ | c. $-\pi \csc \frac{\pi x}{2} \cot \frac{\pi x}{2}$ |
| 16. a. $\sec x \tan x$ | b. $4 \sec 3x \tan 3x$ | c. $\sec \frac{\pi x}{2} \tan \frac{\pi x}{2}$ |

Finding Indefinite Integrals

In Exercises 17–54, find the most general antiderivative or indefinite integral. Check your answers by differentiation.

- | | |
|--|--|
| 17. $\int (x + 1) dx$ | 18. $\int (5 - 6x) dx$ |
| 19. $\int \left(3t^2 + \frac{t}{2}\right) dt$ | 20. $\int \left(\frac{t^2}{2} + 4t^3\right) dt$ |
| 21. $\int (2x^3 - 5x + 7) dx$ | 22. $\int (1 - x^2 - 3x^5) dx$ |
| 23. $\int \left(\frac{1}{x^2} - x^2 - \frac{1}{3}\right) dx$ | 24. $\int \left(\frac{1}{5} - \frac{2}{x^3} + 2x\right) dx$ |
| 25. $\int x^{-1/3} dx$ | 26. $\int x^{-5/4} dx$ |
| 27. $\int (\sqrt{x} + \sqrt[3]{x}) dx$ | 28. $\int \left(\frac{\sqrt{x}}{2} + \frac{2}{\sqrt{x}}\right) dx$ |
| 29. $\int \left(8y - \frac{2}{y^{1/4}}\right) dy$ | 30. $\int \left(\frac{1}{7} - \frac{1}{y^{5/4}}\right) dy$ |
| 31. $\int 2x(1 - x^{-3}) dx$ | 32. $\int x^{-3}(x + 1) dx$ |
| 33. $\int \frac{t\sqrt{t} + \sqrt{t}}{t^2} dt$ | 34. $\int \frac{4 + \sqrt{t}}{t^3} dt$ |
| 35. $\int (-2 \cos t) dt$ | 36. $\int (-5 \sin t) dt$ |
| 37. $\int 7 \sin \frac{\theta}{3} d\theta$ | 38. $\int 3 \cos 5\theta d\theta$ |
| 39. $\int (-3 \csc^2 x) dx$ | 40. $\int \left(-\frac{\sec^2 x}{3}\right) dx$ |
| 41. $\int \frac{\csc \theta \cot \theta}{2} d\theta$ | 42. $\int \frac{2}{5} \sec \theta \tan \theta d\theta$ |

$$43. \int (4 \sec x \tan x - 2 \sec^2 x) dx \quad 44. \int \frac{1}{2} (\csc^2 x - \csc x \cot x) dx$$

$$45. \int (\sin 2x - \csc^2 x) dx \quad 46. \int (2 \cos 2x - 3 \sin 3x) dx$$

$$47. \int \frac{1 + \cos 4t}{2} dt \quad 48. \int \frac{1 - \cos 6t}{2} dt$$

$$49. \int (1 + \tan^2 \theta) d\theta \quad 50. \int (2 + \tan^2 \theta) d\theta$$

(Hint: $1 + \tan^2 \theta = \sec^2 \theta$)

$$51. \int \cot^2 x dx \quad 52. \int (1 - \cot^2 x) dx$$

(Hint: $1 + \cot^2 x = \csc^2 x$)

$$53. \int \cos \theta (\tan \theta + \sec \theta) d\theta \quad 54. \int \frac{\csc \theta}{\csc \theta - \sin \theta} d\theta$$

Checking Antiderivative Formulas

Verify the formulas in Exercises 55–60 by differentiation.

$$55. \int (7x - 2)^3 dx = \frac{(7x - 2)^4}{28} + C$$

$$56. \int (3x + 5)^{-2} dx = -\frac{(3x + 5)^{-1}}{3} + C$$

$$57. \int \sec^2(5x - 1) dx = \frac{1}{5} \tan(5x - 1) + C$$

$$58. \int \csc^2\left(\frac{x-1}{3}\right) dx = -3 \cot\left(\frac{x-1}{3}\right) + C$$

$$59. \int \frac{1}{(x+1)^2} dx = -\frac{1}{x+1} + C$$

$$60. \int \frac{1}{(x+1)^2} dx = \frac{x}{x+1} + C$$

61. Right, or wrong? Say which for each formula and give a brief reason for each answer.

$$\text{a. } \int x \sin x dx = \frac{x^2}{2} \sin x + C$$

$$\text{b. } \int x \sin x dx = -x \cos x + C$$

$$\text{c. } \int x \sin x dx = -x \cos x + \sin x + C$$

62. Right, or wrong? Say which for each formula and give a brief reason for each answer.

$$\text{a. } \int \tan \theta \sec^2 \theta d\theta = \frac{\sec^3 \theta}{3} + C$$

$$\text{b. } \int \tan \theta \sec^2 \theta d\theta = \frac{1}{2} \tan^2 \theta + C$$

$$\text{c. } \int \tan \theta \sec^2 \theta d\theta = \frac{1}{2} \sec^2 \theta + C$$

63. Right, or wrong? Say which for each formula and give a brief reason for each answer.

$$\text{a. } \int (2x + 1)^2 dx = \frac{(2x + 1)^3}{3} + C$$

$$\text{b. } \int 3(2x + 1)^2 dx = (2x + 1)^3 + C$$

$$\text{c. } \int 6(2x + 1)^2 dx = (2x + 1)^3 + C$$

64. Right, or wrong? Say which for each formula and give a brief reason for each answer.

$$\text{a. } \int \sqrt{2x + 1} dx = \sqrt{x^2 + x} + C$$

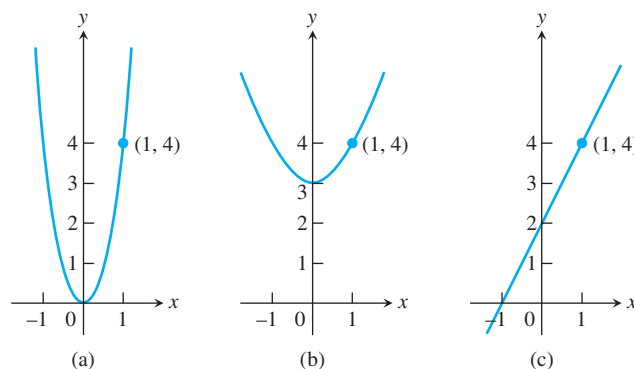
$$\text{b. } \int \sqrt{2x + 1} dx = \sqrt{x^2 + x} + C$$

$$\text{c. } \int \sqrt{2x + 1} dx = \frac{1}{3} (\sqrt{2x + 1})^3 + C$$

Initial Value Problems

65. Which of the following graphs shows the solution of the initial value problem

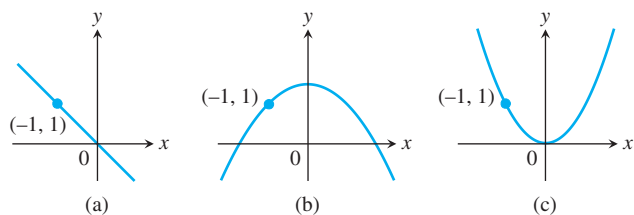
$$\frac{dy}{dx} = 2x, \quad y = 4 \text{ when } x = 1?$$



Give reasons for your answer.

66. Which of the following graphs shows the solution of the initial value problem

$$\frac{dy}{dx} = -x, \quad y = 1 \text{ when } x = -1?$$



Give reasons for your answer.

Solve the initial value problems in Exercises 67–86.

67. $\frac{dy}{dx} = 2x - 7, \quad y(2) = 0$

68. $\frac{dy}{dx} = 10 - x, \quad y(0) = -1$

69. $\frac{dy}{dx} = \frac{1}{x^2} + x, \quad x > 0; \quad y(2) = 1$

70. $\frac{dy}{dx} = 9x^2 - 4x + 5, \quad y(-1) = 0$

71. $\frac{dy}{dx} = 3x^{-2/3}, \quad y(-1) = -5$

72. $\frac{dy}{dx} = \frac{1}{2\sqrt{x}}, \quad y(4) = 0$

73. $\frac{ds}{dt} = 1 + \cos t, \quad s(0) = 4$

74. $\frac{ds}{dt} = \cos t + \sin t, \quad s(\pi) = 1$

75. $\frac{dr}{d\theta} = -\pi \sin \pi\theta, \quad r(0) = 0$

76. $\frac{dr}{d\theta} = \cos \pi\theta, \quad r(0) = 1$

77. $\frac{dv}{dt} = \frac{1}{2} \sec t \tan t, \quad v(0) = 1$

78. $\frac{dv}{dt} = 8t + \csc^2 t, \quad v\left(\frac{\pi}{2}\right) = -7$

79. $\frac{d^2y}{dx^2} = 2 - 6x; \quad y'(0) = 4, \quad y(0) = 1$

80. $\frac{d^2y}{dx^2} = 0; \quad y'(0) = 2, \quad y(0) = 0$

81. $\frac{d^2r}{dt^2} = \frac{2}{t^3}; \quad \left.\frac{dr}{dt}\right|_{t=1} = 1, \quad r(1) = 1$

82. $\frac{d^2s}{dt^2} = \frac{3t}{8}; \quad \left.\frac{ds}{dt}\right|_{t=4} = 3, \quad s(4) = 4$

83. $\frac{d^3y}{dx^3} = 6; \quad y''(0) = -8, \quad y'(0) = 0, \quad y(0) = 5$

84. $\frac{d^3\theta}{dt^3} = 0; \quad \theta''(0) = -2, \quad \theta'(0) = -\frac{1}{2}, \quad \theta(0) = \sqrt{2}$

85. $y^{(4)} = -\sin t + \cos t; \quad y'''(0) = 7, \quad y''(0) = y'(0) = -1, \quad y(0) = 0$

86. $y^{(4)} = -\cos x + 8 \sin 2x; \quad y'''(0) = 0, \quad y''(0) = y'(0) = 1, \quad y(0) = 3$

Finding Curves

87. Find the curve $y = f(x)$ in the xy -plane that passes through the point $(9, 4)$ and whose slope at each point is $3\sqrt{x}$.

88. a. Find a curve $y = f(x)$ with the following properties:

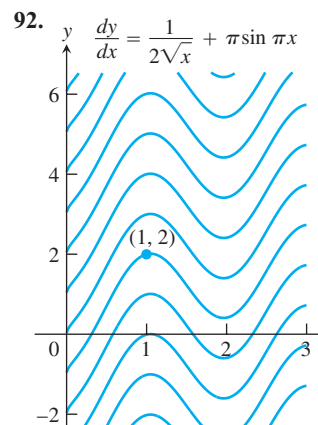
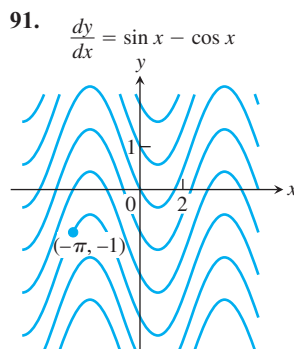
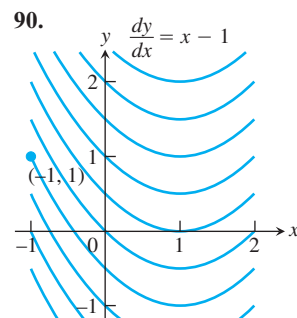
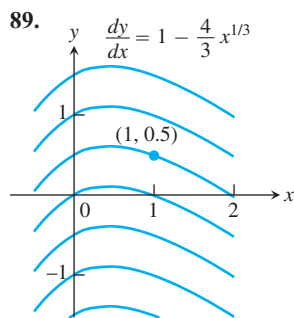
i) $\frac{d^2y}{dx^2} = 6x$

ii) Its graph passes through the point $(0, 1)$, and has a horizontal tangent there.

b. How many curves like this are there? How do you know?

Solution (Integral) Curves

Exercises 89–92 show solution curves of differential equations. In each exercise, find an equation for the curve through the labeled point.



Applications

93. Finding displacement from an antiderivative of velocity

a. Suppose that the velocity of a body moving along the s -axis is

$$\frac{ds}{dt} = v = 9.8t - 3.$$

i) Find the body's displacement over the time interval from $t = 1$ to $t = 3$ given that $s = 5$ when $t = 0$.

ii) Find the body's displacement from $t = 1$ to $t = 3$ given that $s = -2$ when $t = 0$.

iii) Now find the body's displacement from $t = 1$ to $t = 3$ given that $s = s_0$ when $t = 0$.

- b. Suppose that the position s of a body moving along a coordinate line is a differentiable function of time t . Is it true that once you know an antiderivative of the velocity function ds/dt you can find the body's displacement from $t = a$ to $t = b$ even if you do not know the body's exact position at either of those times? Give reasons for your answer.

94. Liftoff from Earth A rocket lifts off the surface of Earth with a constant acceleration of 20 m/sec^2 . How fast will the rocket be going 1 min later?

95. Stopping a car in time You are driving along a highway at a steady 60 mph (88 ft/sec) when you see an accident ahead and slam on the brakes. What constant deceleration is required to stop your car in 242 ft? To find out, carry out the following steps.

1. Solve the initial value problem

$$\text{Differential equation: } \frac{d^2s}{dt^2} = -k \quad (k \text{ constant})$$

$$\text{Initial conditions: } \frac{ds}{dt} = 88 \text{ and } s = 0 \text{ when } t = 0.$$

Measuring time and distance from when the brakes are applied.

2. Find the value of t that makes $ds/dt = 0$. (The answer will involve k .)
3. Find the value of k that makes $s = 242$ for the value of t you found in Step 2.

96. Stopping a motorcycle The State of Illinois Cycle Rider Safety Program requires riders to be able to brake from 30 mph (44 ft/sec) to 0 in 45 ft. What constant deceleration does it take to do that?

97. Motion along a coordinate line A particle moves on a coordinate line with acceleration $a = d^2s/dt^2 = 15\sqrt{t} - (3/\sqrt{t})$, subject to the conditions that $ds/dt = 4$ and $s = 0$ when $t = 1$. Find

- a. the velocity $v = ds/dt$ in terms of t
b. the position s in terms of t .

T 98. The hammer and the feather When *Apollo 15* astronaut David Scott dropped a hammer and a feather on the moon to demonstrate that in a vacuum all bodies fall with the same (constant) acceleration, he dropped them from about 4 ft above the ground. The television footage of the event shows the hammer and the feather falling more slowly than on Earth, where, in a vacuum, they would have taken only half a second to fall the 4 ft. How long did it take the hammer and feather to fall 4 ft on the moon? To find out, solve the following initial value problem for s as a function of t . Then find the value of t that makes s equal to 0.

$$\text{Differential equation: } \frac{d^2s}{dt^2} = -5.2 \text{ ft/sec}^2$$

$$\text{Initial conditions: } \frac{ds}{dt} = 0 \text{ and } s = 4 \text{ when } t = 0$$

99. Motion with constant acceleration The standard equation for the position s of a body moving with a constant acceleration a along a coordinate line is

$$s = \frac{a}{2}t^2 + v_0t + s_0, \quad (1)$$

where v_0 and s_0 are the body's velocity and position at time $t = 0$. Derive this equation by solving the initial value problem

$$\text{Differential equation: } \frac{d^2s}{dt^2} = a$$

$$\text{Initial conditions: } \frac{ds}{dt} = v_0 \text{ and } s = s_0 \text{ when } t = 0.$$

100. Free fall near the surface of a planet For free fall near the surface of a planet where the acceleration due to gravity has a constant magnitude of g length-units/sec², Equation (1) in Exercise 99 takes the form

$$s = -\frac{1}{2}gt^2 + v_0t + s_0, \quad (2)$$

where s is the body's height above the surface. The equation has a minus sign because the acceleration acts downward, in the direction of decreasing s . The velocity v_0 is positive if the object is rising at time $t = 0$ and negative if the object is falling.

Instead of using the result of Exercise 99, you can derive Equation (2) directly by solving an appropriate initial value problem. What initial value problem? Solve it to be sure you have the right one, explaining the solution steps as you go along.

Theory and Examples

101. Suppose that

$$f(x) = \frac{d}{dx}(1 - \sqrt{x}) \quad \text{and} \quad g(x) = \frac{d}{dx}(x + 2).$$

Find:

- | | |
|----------------------------|----------------------------|
| a. $\int f(x) dx$ | b. $\int g(x) dx$ |
| c. $\int [-f(x)] dx$ | d. $\int [-g(x)] dx$ |
| e. $\int [f(x) + g(x)] dx$ | f. $\int [f(x) - g(x)] dx$ |

102. Uniqueness of solutions If differentiable functions $y = F(x)$ and $y = G(x)$ both solve the initial value problem

$$\frac{dy}{dx} = f(x), \quad y(x_0) = y_0,$$

on an interval I , must $F(x) = G(x)$ for every x in I ? Give reasons for your answer.

COMPUTER EXPLORATIONS

Use a CAS to solve the initial problems in Exercises 103–106. Plot the solution curves.

103. $y' = \cos^2 x + \sin x$, $y(\pi) = 1$

104. $y' = \frac{1}{x} + x$, $y(1) = -1$

105. $y' = \frac{1}{\sqrt{4-x^2}}$, $y(0) = 2$

106. $y'' = \frac{2}{x} + \sqrt{x}$, $y(1) = 0$, $y'(1) = 0$

Chapter 4 Additional and Advanced Exercises

- What can you say about a function whose maximum and minimum values on an interval are equal? Give reasons for your answer.
- Is it true that a discontinuous function cannot have both an absolute maximum and an absolute minimum value on a closed interval? Give reasons for your answer.
- Can you conclude anything about the extreme values of a continuous function on an open interval? On a half-open interval? Give reasons for your answer.
- Local extrema** Use the sign pattern for the derivative

$$\frac{df}{dx} = 6(x-1)(x-2)^2(x-3)^3(x-4)^4$$

to identify the points where f has local maximum and minimum values.

5. Local extrema

- a. Suppose that the first derivative of $y = f(x)$ is

$$y' = 6(x+1)(x-2)^2.$$

At what points, if any, does the graph of f have a local maximum, local minimum, or point of inflection?

- b. Suppose that the first derivative of $y = f(x)$ is

$$y' = 6x(x+1)(x-2).$$

At what points, if any, does the graph of f have a local maximum, local minimum, or point of inflection?

- If $f'(x) \leq 2$ for all x , what is the most the values of f can increase on $[0, 6]$? Give reasons for your answer.
- Bounding a function** Suppose that f is continuous on $[a, b]$ and that c is an interior point of the interval. Show that if $f'(x) \leq 0$ on $[a, c]$ and $f'(x) \geq 0$ on $(c, b]$, then $f(x)$ is never less than $f(c)$ on $[a, b]$.
- An inequality**
 - Show that $-1/2 \leq x/(1+x^2) \leq 1/2$ for every value of x .
 - Suppose that f is a function whose derivative is $f'(x) = x/(1+x^2)$. Use the result in part (a) to show that

$$|f(b) - f(a)| \leq \frac{1}{2}|b - a|$$

for any a and b .

- The derivative of $f(x) = x^2$ is zero at $x = 0$, but f is not a constant function. Doesn't this contradict the corollary of the Mean Value Theorem that says that functions with zero derivatives are constant? Give reasons for your answer.
- Extrema and inflection points** Let $h = fg$ be the product of two differentiable functions of x .

- If f and g are positive, with local maxima at $x = a$, and if f' and g' change sign at a , does h have a local maximum at a ?
- If the graphs of f and g have inflection points at $x = a$, does the graph of h have an inflection point at a ?

In either case, if the answer is yes, give a proof. If the answer is no, give a counterexample.

- Finding a function** Use the following information to find the values of a , b , and c in the formula $f(x) = (x+a)/(bx^2+cx+2)$.
 - The values of a , b , and c are either 0 or 1.
 - The graph of f passes through the point $(-1, 0)$.
 - The line $y = 1$ is an asymptote of the graph of f .

- Horizontal tangent** For what value or values of the constant k will the curve $y = x^3 + kx^2 + 3x - 4$ have exactly one horizontal tangent?

- Largest inscribed triangle** Points A and B lie at the ends of a diameter of a unit circle and point C lies on the circumference. Is it true that the area of triangle ABC is largest when the triangle is isosceles? How do you know?

- Proving the second derivative test** The Second Derivative Test for Local Maxima and Minima (Section 4.4) says:

- f has a local maximum value at $x = c$ if $f'(c) = 0$ and $f''(c) < 0$
- f has a local minimum value at $x = c$ if $f'(c) = 0$ and $f''(c) > 0$.

To prove statement (a), let $\epsilon = (1/2)|f''(c)|$. Then use the fact that

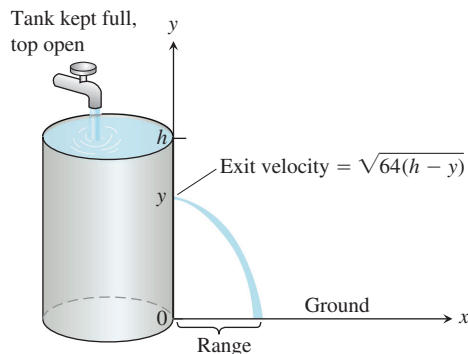
$$f''(c) = \lim_{h \rightarrow 0} \frac{f'(c+h) - f'(c)}{h} = \lim_{h \rightarrow 0} \frac{f'(c+h)}{h}$$

to conclude that for some $\delta > 0$,

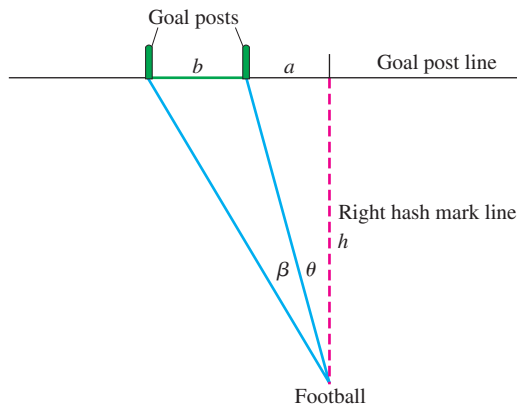
$$0 < |h| < \delta \quad \Rightarrow \quad \frac{f'(c+h)}{h} < f''(c) + \epsilon < 0.$$

Thus, $f'(c+h)$ is positive for $-\delta < h < 0$ and negative for $0 < h < \delta$. Prove statement (b) in a similar way.

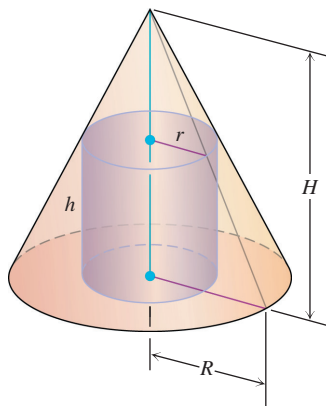
- Hole in a water tank** You want to bore a hole in the side of the tank shown here at a height that will make the stream of water coming out hit the ground as far from the tank as possible. If you drill the hole near the top, where the pressure is low, the water will exit slowly but spend a relatively long time in the air. If you drill the hole near the bottom, the water will exit at a higher velocity but have only a short time to fall. Where is the best place, if any, for the hole? (*Hint:* How long will it take an exiting particle of water to fall from height y to the ground?)



- 16. Kicking a field goal** An American football player wants to kick a field goal with the ball being on a right hash mark. Assume that the goal posts are b feet apart and that the hash mark line is a distance $a > 0$ feet from the right goal post. (See the accompanying figure.) Find the distance h from the goal post line that gives the kicker his largest angle β . Assume that the football field is flat.



- 17. A max-min problem with a variable answer** Sometimes the solution of a max-min problem depends on the proportions of the shapes involved. As a case in point, suppose that a right circular cylinder of radius r and height h is inscribed in a right circular cone of radius R and height H , as shown here. Find the value of r (in terms of R and H) that maximizes the total surface area of the cylinder (including top and bottom). As you will see, the solution depends on whether $H \leq 2R$ or $H > 2R$.



- 18. Minimizing a parameter** Find the smallest value of the positive constant m that will make $mx - 1 + (1/x)$ greater than or equal to zero for all positive values of x .

- 19. Evaluate the following limits.**

$$\begin{array}{ll} \text{a. } \lim_{x \rightarrow 0} \frac{2 \sin 5x}{3x} & \text{b. } \lim_{x \rightarrow 0} \sin 5x \cot 3x \\ \text{c. } \lim_{x \rightarrow 0} x \csc^2 \sqrt{2x} & \text{d. } \lim_{x \rightarrow \pi/2} (\sec x - \tan x) \\ \text{e. } \lim_{x \rightarrow 0} \frac{x - \sin x}{x - \tan x} & \text{f. } \lim_{x \rightarrow 0} \frac{\sin x^2}{x \sin x} \\ \text{g. } \lim_{x \rightarrow 0} \frac{\sec x - 1}{x^2} & \text{h. } \lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4} \end{array}$$

- 20. L'Hôpital's Rule** does not help with the following limits. Find them some other way.

$$\begin{array}{ll} \text{a. } \lim_{x \rightarrow \infty} \frac{\sqrt{x+5}}{\sqrt{x+5}} & \text{b. } \lim_{x \rightarrow \infty} \frac{2x}{x + 7\sqrt{x}} \end{array}$$

- 21. Suppose** that it costs a company $y = a + bx$ dollars to produce x units per week. It can sell x units per week at a price of $P = c - ex$ dollars per unit. Each of a , b , c , and e represents a positive constant. **(a)** What production level maximizes the profit? **(b)** What is the corresponding price? **(c)** What is the weekly profit at this level of production? **(d)** At what price should each item be sold to maximize profits if the government imposes a tax of t dollars per item sold? Comment on the difference between this price and the price before the tax.

- 22. Estimating reciprocals without division** You can estimate the value of the reciprocal of a number a without ever dividing by a if you apply Newton's method to the function $f(x) = (1/x) - a$. For example, if $a = 3$, the function involved is $f(x) = (1/x) - 3$.

- a.** Graph $y = (1/x) - 3$. Where does the graph cross the x -axis?
b. Show that the recursion formula in this case is

$$x_{n+1} = x_n(2 - 3x_n),$$

so there is no need for division.

- 23. To find** $x = \sqrt[q]{a}$, we apply Newton's method to $f(x) = x^q - a$. Here we assume that a is a positive real number and q is a positive integer. Show that x_1 is a "weighted average" of x_0 and a/x_0^{q-1} , and find the coefficients m_0, m_1 such that

$$x_1 = m_0 x_0 + m_1 \left(\frac{a}{x_0^{q-1}} \right), \quad \begin{array}{l} m_0 > 0, m_1 > 0, \\ m_0 + m_1 = 1. \end{array}$$

What conclusion would you reach if x_0 and a/x_0^{q-1} were equal? What would be the value of x_1 in that case?

- 24. The family** of straight lines $y = ax + b$ (a, b arbitrary constants) can be characterized by the relation $y'' = 0$. Find a similar relation satisfied by the family of all circles

$$(x - h)^2 + (y - h)^2 = r^2,$$

where h and r are arbitrary constants. (Hint: Eliminate h and r from the set of three equations including the given one and two obtained by successive differentiation.)

25. Assume that the brakes of an automobile produce a constant deceleration of k ft/sec². (a) Determine what k must be to bring an automobile traveling 60 mi/hr (88 ft/sec) to rest in a distance of 100 ft from the point where the brakes are applied. (b) With the same k , how far would a car traveling 30 mi/hr travel before being brought to a stop?
26. Let $f(x)$, $g(x)$ be two continuously differentiable functions satisfying the relationships $f'(x) = g(x)$ and $f''(x) = -f(x)$. Let $h(x) = f^2(x) + g^2(x)$. If $h(0) = 5$, find $h(10)$.
27. Can there be a curve satisfying the following conditions? d^2y/dx^2 is everywhere equal to zero and, when $x = 0$, $y = 0$ and $dy/dx = 1$. Give a reason for your answer.
28. Find the equation for the curve in the xy -plane that passes through the point $(1, -1)$ if its slope at x is always $3x^2 + 2$.
29. A particle moves along the x -axis. Its acceleration is $a = -t^2$. At $t = 0$, the particle is at the origin. In the course of its motion, it reaches the point $x = b$, where $b > 0$, but no point beyond b . Determine its velocity at $t = 0$.
30. A particle moves with acceleration $a = \sqrt{t} - (1/\sqrt{t})$. Assuming that the velocity $v = 4/3$ and the position $s = -4/15$ when $t = 0$, find

- a. the velocity v in terms of t .
 b. the position s in terms of t .

31. Given $f(x) = ax^2 + 2bx + c$ with $a > 0$. By considering the minimum, prove that $f(x) \geq 0$ for all real x if, and only if, $b^2 - ac \leq 0$.

32. Schwarz's inequality

- a. In Exercise 31, let

$$f(x) = (a_1x + b_1)^2 + (a_2x + b_2)^2 + \cdots + (a_nx + b_n)^2,$$

and deduce Schwarz's inequality:

$$(a_1b_1 + a_2b_2 + \cdots + a_nb_n)^2 \leq (a_1^2 + a_2^2 + \cdots + a_n^2)(b_1^2 + b_2^2 + \cdots + b_n^2).$$

- b. Show that equality holds in Schwarz's inequality only if there exists a real number x that makes a_ix equal $-b_i$ for every value of i from 1 to n .

Chapter 4

Practice Exercises

Existence of Extreme Values

1. Does $f(x) = x^3 + 2x + \tan x$ have any local maximum or minimum values? Give reasons for your answer.
2. Does $g(x) = \csc x + 2 \cot x$ have any local maximum values? Give reasons for your answer.
3. Does $f(x) = (7 + x)(11 - 3x)^{1/3}$ have an absolute minimum value? An absolute maximum? If so, find them or give reasons why they fail to exist. List all critical points of f .

4. Find values of a and b such that the function

$$f(x) = \frac{ax + b}{x^2 - 1}$$

has a local extreme value of 1 at $x = 3$. Is this extreme value a local maximum, or a local minimum? Give reasons for your answer.

5. The greatest integer function $f(x) = \lfloor x \rfloor$, defined for all values of x , assumes a local maximum value of 0 at each point of $[0, 1)$. Could any of these local maximum values also be local minimum values of f ? Give reasons for your answer.
6. a. Give an example of a differentiable function f whose first derivative is zero at some point c even though f has neither a local maximum nor a local minimum at c .
b. How is this consistent with Theorem 2 in Section 4.1? Give reasons for your answer.
7. The function $y = 1/x$ does not take on either a maximum or a minimum on the interval $0 < x < 1$ even though the function is continuous on this interval. Does this contradict the Extreme Value Theorem for continuous functions? Why?
8. What are the maximum and minimum values of the function $y = |x|$ on the interval $-1 \leq x < 1$? Notice that the interval is not closed. Is this consistent with the Extreme Value Theorem for continuous functions? Why?

- T** 9. A graph that is large enough to show a function's global behavior may fail to reveal important local features. The graph of $f(x) = (x^8/8) - (x^6/2) - x^5 + 5x^3$ is a case in point.
- a. Graph f over the interval $-2.5 \leq x \leq 2.5$. Where does the graph appear to have local extreme values or points of inflection?
- b. Now factor $f'(x)$ and show that f has a local maximum at $x = \sqrt[3]{5} \approx 1.70998$ and local minima at $x = \pm\sqrt{3} \approx \pm 1.73205$.
- c. Zoom in on the graph to find a viewing window that shows the presence of the extreme values at $x = \sqrt[3]{5}$ and $x = \sqrt{3}$.

The moral here is that without calculus the existence of two of the three extreme values would probably have gone unnoticed. On any normal graph of the function, the values would lie close enough together to fall within the dimensions of a single pixel on the screen.

(Source: *Uses of Technology in the Mathematics Curriculum*, by Benny Evans and Jerry Johnson, Oklahoma State University, published in 1990 under National Science Foundation Grant USE-8950044.)

- T** 10. (Continuation of Exercise 9.)
- a. Graph $f(x) = (x^8/8) - (2/5)x^5 - 5x - (5/x^2) + 11$ over the interval $-2 \leq x \leq 2$. Where does the graph appear to have local extreme values or points of inflection?
- b. Show that f has a local maximum value at $x = \sqrt[3]{5} \approx 1.2585$ and a local minimum value at $x = \sqrt[3]{2} \approx 1.2599$.
- c. Zoom in to find a viewing window that shows the presence of the extreme values at $x = \sqrt[3]{5}$ and $x = \sqrt[3]{2}$.

The Mean Value Theorem

11. a. Show that $g(t) = \sin^2 t - 3t$ decreases on every interval in its domain.
b. How many solutions does the equation $\sin^2 t - 3t = 5$ have? Give reasons for your answer.
12. a. Show that $y = \tan \theta$ increases on every interval in its domain.
b. If the conclusion in part (a) is really correct, how do you explain the fact that $\tan \pi = 0$ is less than $\tan(\pi/4) = 1$?
13. a. Show that the equation $x^4 + 2x^2 - 2 = 0$ has exactly one solution on $[0, 1]$.
T b. Find the solution to as many decimal places as you can.
14. a. Show that $f(x) = x/(x + 1)$ increases on every interval in its domain.
b. Show that $f(x) = x^3 + 2x$ has no local maximum or minimum values.
15. **Water in a reservoir** As a result of a heavy rain, the volume of water in a reservoir increased by 1400 acre-ft in 24 hours. Show that at some instant during that period the reservoir's volume was increasing at a rate in excess of 225,000 gal/min. (An acre-foot is 43,560 ft³, the volume that would cover 1 acre to the depth of 1 ft. A cubic foot holds 7.48 gal.)
16. The formula $F(x) = 3x + C$ gives a different function for each value of C . All of these functions, however, have the same derivative with respect to x , namely $F'(x) = 3$. Are these the only differentiable functions whose derivative is 3? Could there be any others? Give reasons for your answers.
17. Show that

$$\frac{d}{dx} \left(\frac{x}{x+1} \right) = \frac{d}{dx} \left(-\frac{1}{x+1} \right)$$

even though

$$\frac{x}{x+1} \neq -\frac{1}{x+1}.$$

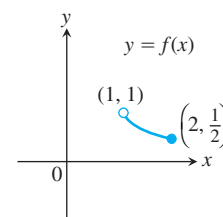
Doesn't this contradict Corollary 2 of the Mean Value Theorem? Give reasons for your answer.

18. Calculate the first derivatives of $f(x) = x^2/(x^2 + 1)$ and $g(x) = -1/(x^2 + 1)$. What can you conclude about the graphs of these functions?

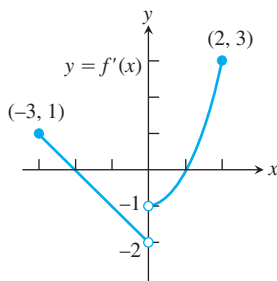
Conclusions from Graphs

In Exercises 19 and 20, use the graph to answer the questions.

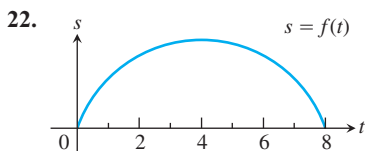
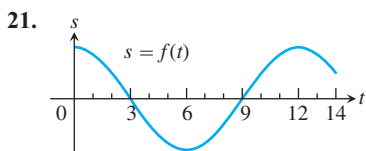
19. Identify any global extreme values of f and the values of x at which they occur.



20. Estimate the intervals on which the function $y = f(x)$ is
- increasing.
 - decreasing.
 - Use the given graph of f' to indicate where any local extreme values of the function occur, and whether each extreme is a relative maximum or minimum.



Each of the graphs in Exercises 21 and 22 is the graph of the position function $s = f(t)$ of a body moving on a coordinate line (t represents time). At approximately what times (if any) is each body's (a) velocity equal to zero? (b) Acceleration equal to zero? During approximately what time intervals does the body move (c) forward? (d) Backward?



Graphs and Graphing

Graph the curves in Exercises 23–32.

- $y = x^2 - (x^3/6)$
- $y = x^3 - 3x^2 + 3$
- $y = -x^3 + 6x^2 - 9x + 3$
- $y = (1/8)(x^3 + 3x^2 - 9x - 27)$
- $y = x^3(8 - x)$
- $y = x^2(2x^2 - 9)$
- $y = x - 3x^{2/3}$
- $y = x^{1/3}(x - 4)$
- $y = x\sqrt{3 - x}$
- $y = x\sqrt{4 - x^2}$

Each of Exercises 33–38 gives the first derivative of a function $y = f(x)$. (a) At what points, if any, does the graph of f have a local maximum, local minimum, or inflection point? (b) Sketch the general shape of the graph.

- $y' = 16 - x^2$
- $y' = x^2 - x - 6$

- $y' = 6x(x + 1)(x - 2)$
- $y' = x^2(6 - 4x)$
- $y' = x^4 - 2x^2$
- $y' = 4x^2 - x^4$

In Exercises 39–42, graph each function. Then use the function's first derivative to explain what you see.

- $y = x^{2/3} + (x - 1)^{1/3}$
- $y = x^{2/3} + (x - 1)^{2/3}$
- $y = x^{1/3} + (x - 1)^{1/3}$
- $y = x^{2/3} - (x - 1)^{1/3}$

Sketch the graphs of the functions in Exercises 43–50.

- $y = \frac{x + 1}{x - 3}$
- $y = \frac{2x}{x + 5}$
- $y = \frac{x^2 + 1}{x}$
- $y = \frac{x^2 - x + 1}{x}$
- $y = \frac{x^3 + 2}{2x}$
- $y = \frac{x^4 - 1}{x^2}$
- $y = \frac{x^2 - 4}{x^2 - 3}$
- $y = \frac{x^2}{x^2 - 4}$

Applying l'Hôpital's Rule

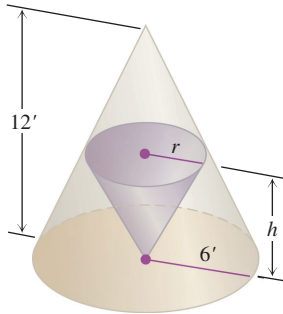
Use l'Hôpital's Rule to find the limits in Exercises 51–62.

- $\lim_{x \rightarrow 1} \frac{x^2 + 3x - 4}{x - 1}$
- $\lim_{x \rightarrow 1} \frac{x^a - 1}{x^b - 1}$
- $\lim_{x \rightarrow \pi} \frac{\tan x}{x}$
- $\lim_{x \rightarrow 0} \frac{\tan x}{x + \sin x}$
- $\lim_{x \rightarrow 0} \frac{\sin^2 x}{\tan(x^2)}$
- $\lim_{x \rightarrow 0} \frac{\sin mx}{\sin nx}$
- $\lim_{x \rightarrow \pi/2^-} \sec 7x \cos 3x$
- $\lim_{x \rightarrow 0^+} \sqrt{x} \sec x$
- $\lim_{x \rightarrow 0} (\csc x - \cot x)$
- $\lim_{x \rightarrow 0} \left(\frac{1}{x^4} - \frac{1}{x^2} \right)$
- $\lim_{x \rightarrow \infty} \left(\sqrt{x^2 + x + 1} - \sqrt{x^2 - x} \right)$
- $\lim_{x \rightarrow \infty} \left(\frac{x^3}{x^2 - 1} - \frac{x^3}{x^2 + 1} \right)$

Optimization

- The sum of two nonnegative numbers is 36. Find the numbers if
 - the difference of their square roots is to be as large as possible.
 - the sum of their square roots is to be as large as possible.
- The sum of two nonnegative numbers is 20. Find the numbers
 - if the product of one number and the square root of the other is to be as large as possible.
 - if one number plus the square root of the other is to be as large as possible.
- An isosceles triangle has its vertex at the origin and its base parallel to the x -axis with the vertices above the axis on the curve $y = 27 - x^2$. Find the largest area the triangle can have.

66. A customer has asked you to design an open-top rectangular stainless steel vat. It is to have a square base and a volume of 32 ft^3 , to be welded from quarter-inch plate, and to weigh no more than necessary. What dimensions do you recommend?
67. Find the height and radius of the largest right circular cylinder that can be put in a sphere of radius $\sqrt{3}$.
68. The figure here shows two right circular cones, one upside down inside the other. The two bases are parallel, and the vertex of the smaller cone lies at the center of the larger cone's base. What values of r and h will give the smaller cone the largest possible volume?



69. **Manufacturing tires** Your company can manufacture x hundred grade A tires and y hundred grade B tires a day, where $0 \leq x \leq 4$ and

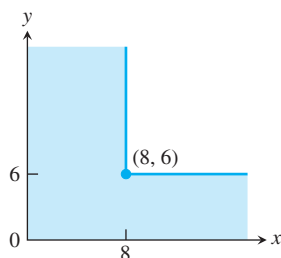
$$y = \frac{40 - 10x}{5 - x}.$$

Your profit on a grade A tire is twice your profit on a grade B tire. What is the most profitable number of each kind to make?

70. **Particle motion** The positions of two particles on the s -axis are $s_1 = \cos t$ and $s_2 = \cos(t + \pi/4)$.
- What is the farthest apart the particles ever get?
 - When do the particles collide?

- T 71. Open-top box** An open-top rectangular box is constructed from a 10-in.-by-16-in. piece of cardboard by cutting squares of equal side length from the corners and folding up the sides. Find analytically the dimensions of the box of largest volume and the maximum volume. Support your answers graphically.

72. **The ladder problem** What is the approximate length (in feet) of the longest ladder you can carry horizontally around the corner of the corridor shown here? Round your answer down to the nearest foot.



Newton's Method

73. Let $f(x) = 3x - x^3$. Show that the equation $f(x) = -4$ has a solution in the interval $[2, 3]$ and use Newton's method to find it.
74. Let $f(x) = x^4 - x^3$. Show that the equation $f(x) = 75$ has a solution in the interval $[3, 4]$ and use Newton's method to find it.

Finding Indefinite Integrals

Find the indefinite integrals (most general antiderivatives) in Exercises 75–90. Check your answers by differentiation.

75. $\int (x^3 + 5x - 7) dx$
76. $\int \left(8t^3 - \frac{t^2}{2} + t \right) dt$
77. $\int \left(3\sqrt{t} + \frac{4}{t^2} \right) dt$
78. $\int \left(\frac{1}{2\sqrt{t}} - \frac{3}{t^4} \right) dt$
79. $\int \frac{dr}{(r + 5)^2}$
80. $\int \frac{6 dr}{(r - \sqrt{2})^3}$
81. $\int 3\theta\sqrt{\theta^2 + 1} d\theta$
82. $\int \frac{\theta}{\sqrt{7 + \theta^2}} d\theta$
83. $\int x^3(1 + x^4)^{-1/4} dx$
84. $\int (2 - x)^{3/5} dx$
85. $\int \sec^2 \frac{s}{10} ds$
86. $\int \csc^2 \pi s ds$
87. $\int \csc \sqrt{2}\theta \cot \sqrt{2}\theta d\theta$
88. $\int \sec \frac{\theta}{3} \tan \frac{\theta}{3} d\theta$
89. $\int \sin^2 \frac{x}{4} dx$
90. $\int \cos^2 \frac{x}{2} dx$ (Hint: $\cos^2 \theta = \frac{1 + \cos 2\theta}{2}$)

Initial Value Problems

Solve the initial value problems in Exercises 91–94.

91. $\frac{dy}{dx} = \frac{x^2 + 1}{x^2}, \quad y(1) = -1$
92. $\frac{dy}{dx} = \left(x + \frac{1}{x} \right)^2, \quad y(1) = 1$
93. $\frac{d^2r}{dt^2} = 15\sqrt{t} + \frac{3}{\sqrt{t}}; \quad r'(1) = 8, \quad r(1) = 0$
94. $\frac{d^3r}{dt^3} = -\cos t; \quad r''(0) = r'(0) = 0, \quad r(0) = -1$

Chapter 4

Questions to Guide Your Review

1. What can be said about the extreme values of a function that is continuous on a closed interval?
2. What does it mean for a function to have a local extreme value on its domain? An absolute extreme value? How are local and absolute extreme values related, if at all? Give examples.
3. How do you find the absolute extrema of a continuous function on a closed interval? Give examples.
4. What are the hypotheses and conclusion of Rolle's Theorem? Are the hypotheses really necessary? Explain.
5. What are the hypotheses and conclusion of the Mean Value Theorem? What physical interpretations might the theorem have?
6. State the Mean Value Theorem's three corollaries.
7. How can you sometimes identify a function $f(x)$ by knowing f' and knowing the value of f at a point $x = x_0$? Give an example.
8. What is the First Derivative Test for Local Extreme Values? Give examples of how it is applied.
9. How do you test a twice-differentiable function to determine where its graph is concave up or concave down? Give examples.
10. What is an inflection point? Give an example. What physical significance do inflection points sometimes have?
11. What is the Second Derivative Test for Local Extreme Values? Give examples of how it is applied.
12. What do the derivatives of a function tell you about the shape of its graph?
13. List the steps you would take to graph a polynomial function. Illustrate with an example.
14. What is a cusp? Give examples.
15. List the steps you would take to graph a rational function. Illustrate with an example.
16. Outline a general strategy for solving max-min problems. Give examples.
17. Describe l'Hôpital's Rule. How do you know when to use the rule and when to stop? Give an example.
18. How can you sometimes handle limits that lead to indeterminate forms ∞/∞ , $\infty \cdot 0$, and $\infty - \infty$. Give examples.
19. Describe Newton's method for solving equations. Give an example. What is the theory behind the method? What are some of the things to watch out for when you use the method?
20. Can a function have more than one antiderivative? If so, how are the antiderivatives related? Explain.
21. What is an indefinite integral? How do you evaluate one? What general formulas do you know for finding indefinite integrals?
22. How can you sometimes solve a differential equation of the form $dy/dx = f(x)$?
23. What is an initial value problem? How do you solve one? Give an example.
24. If you know the acceleration of a body moving along a coordinate line as a function of time, what more do you need to know to find the body's position function? Give an example.

Chapter 4 Technology Application Projects

Mathematica/Maple Module

Motion Along a Straight Line: Position $\rightarrow$ Velocity $\rightarrow$ Acceleration

You will observe the shape of a graph through dramatic animated visualizations of the derivative relations among the position, velocity, and acceleration. Figures in the text can be animated.

Mathematica/Maple Module

Newton's Method: Estimate π to How Many Places?

Plot a function, observe a root, pick a starting point near the root, and use Newton's Iteration Procedure to approximate the root to a desired accuracy. The numbers π , e , and $\sqrt{2}$ are approximated.

INTEGRATION

OVERVIEW One of the great achievements of classical geometry was to obtain formulas for the areas and volumes of triangles, spheres, and cones. In this chapter we study a method to calculate the areas and volumes of these and other more general shapes. The method we develop, called *integration*, is a tool for calculating much more than areas and volumes. The *integral* has many applications in statistics, economics, the sciences, and engineering. It allows us to calculate quantities ranging from probabilities and averages to energy consumption and the forces against a dam's floodgates.

The idea behind integration is that we can effectively compute many quantities by breaking them into small pieces, and then summing the contributions from each small part. We develop the theory of the integral in the setting of area, where it most clearly reveals its nature. We begin with examples involving finite sums. These lead naturally to the question of what happens when more and more terms are summed. Passing to the limit, as the number of terms goes to infinity, then gives an integral. While integration and differentiation are closely connected, we will not see the roles of the derivative and antiderivative emerge until Section 5.4. The nature of their connection, contained in the Fundamental Theorem of Calculus, is one of the most important ideas in calculus.

5.1

Estimating with Finite Sums

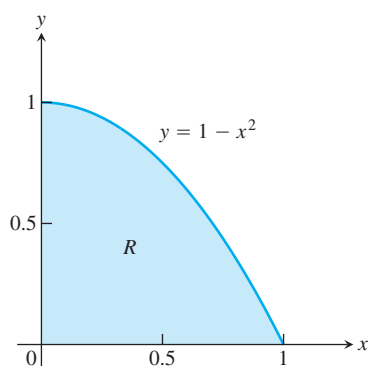


FIGURE 5.1 The area of the region R cannot be found by a simple geometry formula (Example 1).

This section shows how area, average values, and the distance traveled by an object over time can all be approximated by finite sums. Finite sums are the basis for defining the integral in Section 5.3.

Area

The area of a region with a curved boundary can be approximated by summing the areas of a collection of rectangles. Using more rectangles can increase the accuracy of the approximation.

EXAMPLE 1 Approximating Area

What is the area of the shaded region R that lies above the x -axis, below the graph of $y = 1 - x^2$, and between the vertical lines $x = 0$ and $x = 1$? (See Figure 5.1.) An architect might want to know this area to calculate the weight of a custom window with a shape described by R . Unfortunately, there is no simple geometric formula for calculating the areas of shapes having curved boundaries like the region R .

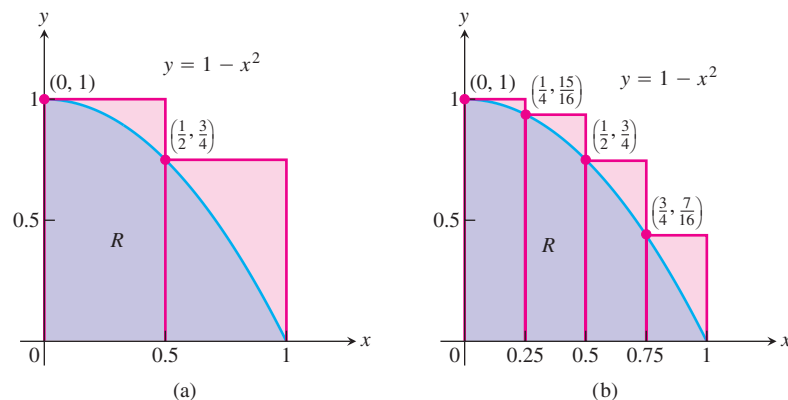


FIGURE 5.2 (a) We get an upper estimate of the area of R by using two rectangles containing R . (b) Four rectangles give a better upper estimate. Both estimates overshoot the true value for the area.

While we do not yet have a method for determining the exact area of R , we can approximate it in a simple way. Figure 5.2a shows two rectangles that together contain the region R . Each rectangle has width $1/2$ and they have heights 1 and $3/4$, moving from left to right. The height of each rectangle is the maximum value of the function f , obtained by evaluating f at the left endpoint of the subinterval of $[0, 1]$ forming the base of the rectangle. The total area of the two rectangles approximates the area A of the region R ,

$$A \approx 1 \cdot \frac{1}{2} + \frac{3}{4} \cdot \frac{1}{2} = \frac{7}{8} = 0.875.$$

This estimate is larger than the true area A , since the two rectangles contain R . We say that 0.875 is an **upper sum** because it is obtained by taking the height of each rectangle as the maximum (uppermost) value of $f(x)$ for x a point in the base interval of the rectangle. In Figure 5.2b, we improve our estimate by using four thinner rectangles, each of width $1/4$, which taken together contain the region R . These four rectangles give the approximation

$$A \approx 1 \cdot \frac{1}{4} + \frac{15}{16} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4} + \frac{7}{16} \cdot \frac{1}{4} = \frac{25}{32} = 0.78125,$$

which is still greater than A since the four rectangles contain R .

Suppose instead we use four rectangles contained *inside* the region R to estimate the area, as in Figure 5.3a. Each rectangle has width $1/4$ as before, but the rectangles are shorter and lie entirely beneath the graph of f . The function $f(x) = 1 - x^2$ is decreasing on $[0, 1]$, so the height of each of these rectangles is given by the value of f at the right endpoint of the subinterval forming its base. The fourth rectangle has zero height and therefore contributes no area. Summing these rectangles with heights equal to the minimum value of $f(x)$ for x a point in each base subinterval, gives a **lower sum** approximation to the area,

$$A \approx \frac{15}{16} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{1}{4} + \frac{7}{16} \cdot \frac{1}{4} + 0 \cdot \frac{1}{4} = \frac{17}{32} = 0.53125.$$

This estimate is smaller than the area A since the rectangles all lie inside of the region R . The true value of A lies somewhere between these lower and upper sums:

$$0.53125 < A < 0.78125.$$

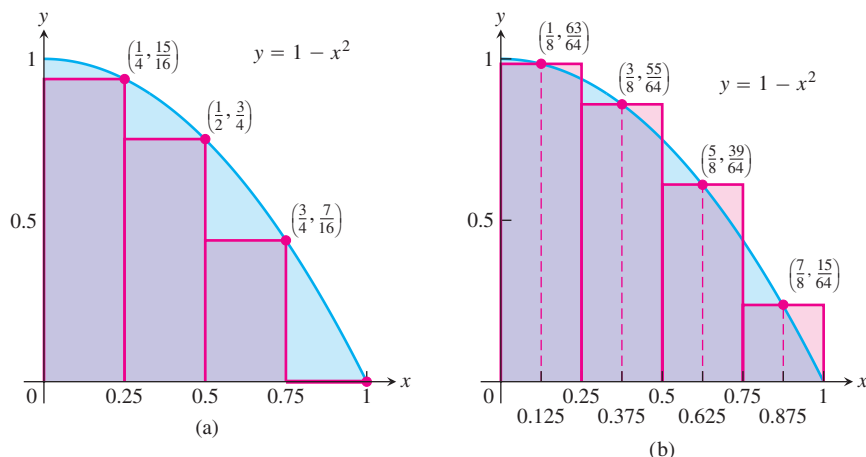


FIGURE 5.3 (a) Rectangles contained in R give an estimate for the area that undershoots the true value. (b) The midpoint rule uses rectangles whose height is the value of $y = f(x)$ at the midpoints of their bases.

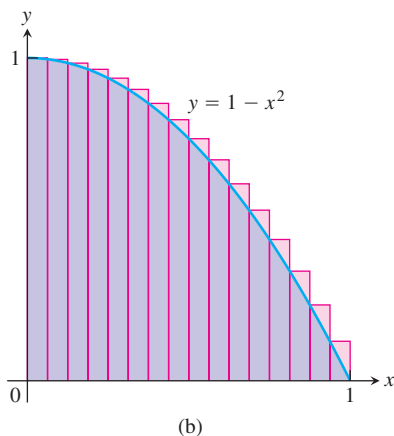
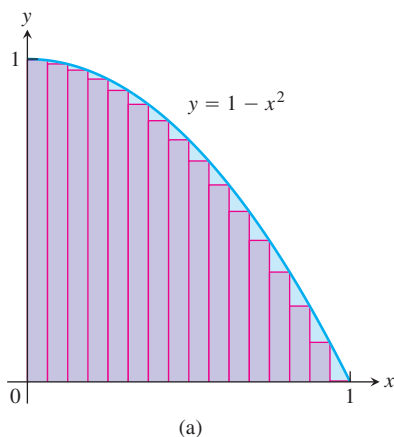


FIGURE 5.4 (a) A lower sum using 16 rectangles of equal width $\Delta x = 1/16$. (b) An upper sum using 16 rectangles.

By considering both lower and upper sum approximations we get not only estimates for the area, but also a bound on the size of the possible error in these estimates since the true value of the area lies somewhere between them. Here the error cannot be greater than the difference $0.78125 - 0.53125 = 0.25$.

Yet another estimate can be obtained by using rectangles whose heights are the values of f at the midpoints of their bases (Figure 5.3b). This method of estimation is called the **midpoint rule** for approximating the area. The midpoint rule gives an estimate that is between a lower sum and an upper sum, but it is not clear whether it overestimates or underestimates the true area. With four rectangles of width $1/4$ as before, the midpoint rule estimates the area of R to be

$$A \approx \frac{63}{64} \cdot \frac{1}{4} + \frac{55}{64} \cdot \frac{1}{4} + \frac{39}{64} \cdot \frac{1}{4} + \frac{15}{64} \cdot \frac{1}{4} = \frac{172}{64} \cdot \frac{1}{4} = 0.671875.$$

In each of our computed sums, the interval $[a, b]$ over which the function f is defined was subdivided into n subintervals of equal width (also called length) $\Delta x = (b - a)/n$, and f was evaluated at a point in each subinterval: c_1 in the first subinterval, c_2 in the second subinterval, and so on. The finite sums then all take the form

$$f(c_1) \Delta x + f(c_2) \Delta x + f(c_3) \Delta x + \cdots + f(c_n) \Delta x.$$

By taking more and more rectangles, with each rectangle thinner than before, it appears that these finite sums give better and better approximations to the true area of the region R .

Figure 5.4a shows a lower sum approximation for the area of R using 16 rectangles of equal width. The sum of their areas is 0.634765625 , which appears close to the true area, but is still smaller since the rectangles lie inside R .

Figure 5.4b shows an upper sum approximation using 16 rectangles of equal width. The sum of their areas is 0.697265625 , which is somewhat larger than the true area because the rectangles taken together contain R . The midpoint rule for 16 rectangles gives a total area approximation of 0.6669921875 , but it is not immediately clear whether this estimate is larger or smaller than the true area.

TABLE 5.1 Finite approximations for the area of R

Number of subintervals	Lower sum	Midpoint rule	Upper sum
2	.375	.6875	.875
4	.53125	.671875	.78125
16	.634765625	.6669921875	.697265625
50	.6566	.6667	.6766
100	.66165	.666675	.67165
1000	.6661665	.6666675	.6671665

Table 5.1 shows the values of upper and lower sum approximations to the area of R using up to 1000 rectangles. In Section 5.2 we will see how to get an exact value of the areas of regions such as R by taking a limit as the base width of each rectangle goes to zero and the number of rectangles goes to infinity. With the techniques developed there, we will be able to show that the area of R is exactly $2/3$.

Distance Traveled

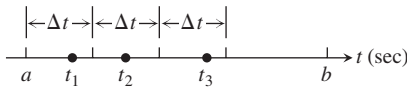
Suppose we know the velocity function $v(t)$ of a car moving down a highway, without changing direction, and want to know how far it traveled between times $t = a$ and $t = b$. If we already know an antiderivative $F(t)$ of $v(t)$ we can find the car's position function $s(t)$ by setting $s(t) = F(t) + C$. The distance traveled can then be found by calculating the change in position, $s(b) - s(a)$ (see Exercise 93, Section 4.8). If the velocity function is determined by recording a speedometer reading at various times on the car, then we have no formula from which to obtain an antiderivative function for velocity. So what do we do in this situation?

When we don't know an antiderivative for the velocity function $v(t)$, we can approximate the distance traveled in the following way. Subdivide the interval $[a, b]$ into short time intervals on each of which the velocity is considered to be fairly constant. Then approximate the distance traveled on each time subinterval with the usual distance formula

$$\text{distance} = \text{velocity} \times \text{time}$$

and add the results across $[a, b]$.

Suppose the subdivided interval looks like



with the subintervals all of equal length Δt . Pick a number t_1 in the first interval. If Δt is so small that the velocity barely changes over a short time interval of duration Δt , then the distance traveled in the first time interval is about $v(t_1) \Delta t$. If t_2 is a number in the second interval, the distance traveled in the second time interval is about $v(t_2) \Delta t$. The sum of the distances traveled over all the time intervals is

$$D \approx v(t_1) \Delta t + v(t_2) \Delta t + \cdots + v(t_n) \Delta t,$$

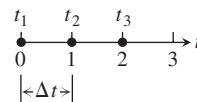
where n is the total number of subintervals.

EXAMPLE 2 Estimating the Height of a Projectile

The velocity function of a projectile fired straight into the air is $f(t) = 160 - 9.8t$ m/sec. Use the summation technique just described to estimate how far the projectile rises during the first 3 sec. How close do the sums come to the exact figure of 435.9 m?

Solution We explore the results for different numbers of intervals and different choices of evaluation points. Notice that $f(t)$ is decreasing, so choosing left endpoints gives an upper sum estimate; choosing right endpoints gives a lower sum estimate.

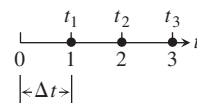
(a) *Three subintervals of length 1, with f evaluated at left endpoints giving an upper sum:*



With f evaluated at $t = 0, 1$, and 2 , we have

$$\begin{aligned} D &\approx f(t_1) \Delta t + f(t_2) \Delta t + f(t_3) \Delta t \\ &= [160 - 9.8(0)](1) + [160 - 9.8(1)](1) + [160 - 9.8(2)](1) \\ &= 450.6. \end{aligned}$$

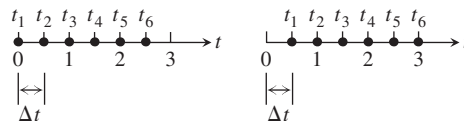
(b) *Three subintervals of length 1, with f evaluated at right endpoints giving a lower sum:*



With f evaluated at $t = 1, 2$, and 3 , we have

$$\begin{aligned} D &\approx f(t_1) \Delta t + f(t_2) \Delta t + f(t_3) \Delta t \\ &= [160 - 9.8(1)](1) + [160 - 9.8(2)](1) + [160 - 9.8(3)](1) \\ &= 421.2. \end{aligned}$$

(c) *With six subintervals of length 1/2, we get*



An upper sum using left endpoints: $D \approx 443.25$; a lower sum using right endpoints: $D \approx 428.55$.

These six-interval estimates are somewhat closer than the three-interval estimates. The results improve as the subintervals get shorter.

As we can see in Table 5.2, the left-endpoint upper sums approach the true value 435.9 from above, whereas the right-endpoint lower sums approach it from below. The true

TABLE 5.2 Travel-distance estimates

Number of subintervals	Length of each subinterval	Upper sum	Lower sum
3	1	450.6	421.2
6	1/2	443.25	428.55
12	1/4	439.57	432.22
24	1/8	437.74	434.06
48	1/16	436.82	434.98
96	1/32	436.36	435.44
192	1/64	436.13	435.67

value lies between these upper and lower sums. The magnitude of the error in the closest entries is 0.23, a small percentage of the true value.

$$\begin{aligned}\text{Error magnitude} &= |\text{true value} - \text{calculated value}| \\ &= |435.9 - 435.67| = 0.23.\end{aligned}$$

$$\text{Error percentage} = \frac{0.23}{435.9} \approx 0.05\%.$$

It would be reasonable to conclude from the table's last entries that the projectile rose about 436 m during its first 3 sec of flight. ■

Displacement Versus Distance Traveled

If a body with position function $s(t)$ moves along a coordinate line without changing direction, we can calculate the total distance it travels from $t = a$ to $t = b$ by summing the distance traveled over small intervals, as in Example 2. If the body changes direction one or more times during the trip, then we need to use the body's *speed* $|v(t)|$, which is the absolute value of its velocity function, $v(t)$, to find the total distance traveled. Using the velocity itself, as in Example 2, only gives an estimate to the body's **displacement**, $s(b) - s(a)$, the difference between its initial and final positions.

To see why, partition the time interval $[a, b]$ into small enough equal subintervals Δt so that the body's velocity does not change very much from time t_{k-1} to t_k . Then $v(t_k)$ gives a good approximation of the velocity throughout the interval. Accordingly, the change in the body's position coordinate during the time interval is about

$$v(t_k) \Delta t.$$

The change is positive if $v(t_k)$ is positive and negative if $v(t_k)$ is negative.

In either case, the distance traveled during the subinterval is about

$$|v(t_k)| \Delta t.$$

The **total distance traveled** is approximately the sum

$$|v(t_1)| \Delta t + |v(t_2)| \Delta t + \cdots + |v(t_n)| \Delta t.$$

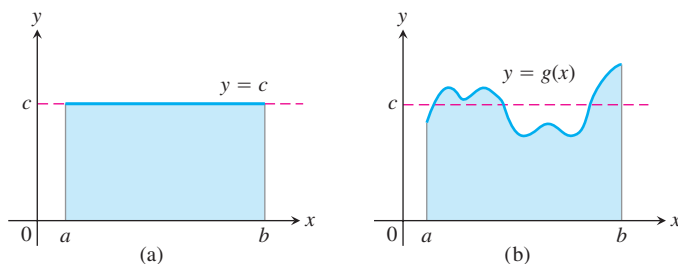


FIGURE 5.5 (a) The average value of $f(x) = c$ on $[a, b]$ is the area of the rectangle divided by $b - a$. (b) The average value of $g(x)$ on $[a, b]$ is the area beneath its graph divided by $b - a$.

Average Value of a Nonnegative Function

The average value of a collection of n numbers $x_1, x_2, \dots, x_n$ is obtained by adding them together and dividing by n . But what is the average value of a continuous function f on an interval $[a, b]$? Such a function can assume infinitely many values. For example, the temperature at a certain location in a town is a continuous function that goes up and down each day. What does it mean to say that the average temperature in the town over the course of a day is 73 degrees?

When a function is constant, this question is easy to answer. A function with constant value c on an interval $[a, b]$ has average value c . When c is positive, its graph over $[a, b]$ gives a rectangle of height c . The average value of the function can then be interpreted geometrically as the area of this rectangle divided by its width $b - a$ (Figure 5.5a).

What if we want to find the average value of a nonconstant function, such as the function g in Figure 5.5b? We can think of this graph as a snapshot of the height of some water that is sloshing around in a tank, between enclosing walls at $x = a$ and $x = b$. As the water moves, its height over each point changes, but its average height remains the same. To get the average height of the water, we let it settle down until it is level and its height is constant. The resulting height c equals the area under the graph of g divided by $b - a$. We are led to *define* the average value of a nonnegative function on an interval $[a, b]$ to be the area under its graph divided by $b - a$. For this definition to be valid, we need a precise understanding of what is meant by the area under a graph. This will be obtained in Section 5.3, but for now we look at two simple examples.

EXAMPLE 3 The Average Value of a Linear Function

What is the average value of the function $f(x) = 3x$ on the interval $[0, 2]$?

Solution The average equals the area under the graph divided by the width of the interval. In this case we do not need finite approximation to estimate the area of the region under the graph: a triangle of height 6 and base 2 has area 6 (Figure 5.6). The width of the interval is $b - a = 2 - 0 = 2$. The average value of the function is $6/2 = 3$. ■

EXAMPLE 4 The Average Value of $\sin x$

Estimate the average value of the function $f(x) = \sin x$ on the interval $[0, \pi]$.

Solution Looking at the graph of $\sin x$ between 0 and π in Figure 5.7, we can see that its average height is somewhere between 0 and 1. To find the average we need to

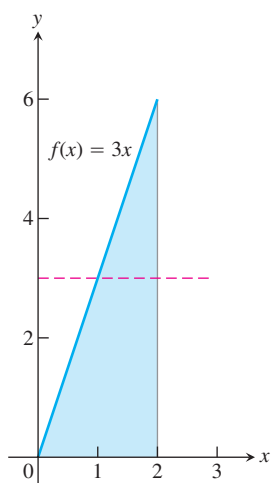


FIGURE 5.6 The average value of $f(x) = 3x$ over $[0, 2]$ is 3 (Example 3).

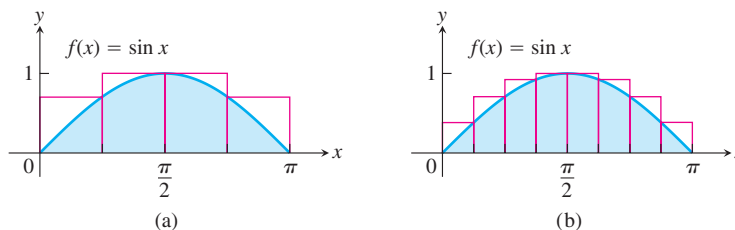


FIGURE 5.7 Approximating the area under $f(x) = \sin x$ between 0 and π to compute the average value of $\sin x$ over $[0, \pi]$, using (a) four rectangles; (b) eight rectangles (Example 4).

calculate the area A under the graph and then divide this area by the length of the interval, $\pi - 0 = \pi$.

We do not have a simple way to determine the area, so we approximate it with finite sums. To get an upper sum estimate, we add the areas of four rectangles of equal width $\pi/4$ that together contain the region beneath the graph of $y = \sin x$ and above the x -axis on $[0, \pi]$. We choose the heights of the rectangles to be the largest value of $\sin x$ on each subinterval. Over a particular subinterval, this largest value may occur at the left endpoint, the right endpoint, or somewhere between them. We evaluate $\sin x$ at this point to get the height of the rectangle for an upper sum. The sum of the rectangle areas then estimates the total area (Figure 5.7a):

$$\begin{aligned} A &\approx \left(\sin \frac{\pi}{4}\right) \cdot \frac{\pi}{4} + \left(\sin \frac{\pi}{2}\right) \cdot \frac{\pi}{4} + \left(\sin \frac{\pi}{2}\right) \cdot \frac{\pi}{4} + \left(\sin \frac{3\pi}{4}\right) \cdot \frac{\pi}{4} \\ &= \left(\frac{1}{\sqrt{2}} + 1 + 1 + \frac{1}{\sqrt{2}}\right) \cdot \frac{\pi}{4} \approx (3.42) \cdot \frac{\pi}{4} \approx 2.69. \end{aligned}$$

To estimate the average value of $\sin x$ we divide the estimated area by π and obtain the approximation $2.69/\pi \approx 0.86$.

If we use eight rectangles of equal width $\pi/8$ all lying above the graph of $y = \sin x$ (Figure 5.7b), we get the area estimate

$$\begin{aligned} A &\approx \left(\sin \frac{\pi}{8} + \sin \frac{\pi}{4} + \sin \frac{3\pi}{8} + \sin \frac{\pi}{2} + \sin \frac{\pi}{2} + \sin \frac{5\pi}{8} + \sin \frac{3\pi}{4} + \sin \frac{7\pi}{8}\right) \cdot \frac{\pi}{8} \\ &\approx (.38 + .71 + .92 + 1 + 1 + .92 + .71 + .38) \cdot \frac{\pi}{8} = (6.02) \cdot \frac{\pi}{8} \approx 2.365. \end{aligned}$$

Dividing this result by the length π of the interval gives a more accurate estimate of 0.753 for the average. Since we used an upper sum to approximate the area, this estimate is still greater than the actual average value of $\sin x$ over $[0, \pi]$. If we use more and more rectangles, with each rectangle getting thinner and thinner, we get closer and closer to the true average value. Using the techniques of Section 5.3, we will show that the true average value is $2/\pi \approx 0.64$.

As before, we could just as well have used rectangles lying under the graph of $y = \sin x$ and calculated a lower sum approximation, or we could have used the midpoint rule. In Section 5.3, we will see that it doesn't matter whether our approximating rectangles are chosen to give upper sums, lower sums, or a sum in between. In each case, the approximations are close to the true area if all the rectangles are sufficiently thin. ■

Summary

The area under the graph of a positive function, the distance traveled by a moving object that doesn't change direction, and the average value of a nonnegative function over an interval can all be approximated by finite sums. First we subdivide the interval into subintervals, treating the appropriate function f as if it were constant over each particular subinterval. Then we multiply the width of each subinterval by the value of f at some point within it, and add these products together. If the interval $[a, b]$ is subdivided into n subintervals of equal widths $\Delta x = (b - a)/n$, and if $f(c_k)$ is the value of f at the chosen point c_k in the k th subinterval, this process gives a finite sum of the form

$$f(c_1) \Delta x + f(c_2) \Delta x + f(c_3) \Delta x + \cdots + f(c_n) \Delta x.$$

The choices for the c_k could maximize or minimize the value of f in the k th subinterval, or give some value in between. The true value lies somewhere between the approximations given by upper sums and lower sums. The finite sum approximations we looked at improved as we took more subintervals of thinner width.

EXERCISES 5.1

Area

In Exercises 1–4 use finite approximations to estimate the area under the graph of the function using

- a lower sum with two rectangles of equal width.
 - a lower sum with four rectangles of equal width.
 - an upper sum with two rectangles of equal width.
 - an upper sum with four rectangles of equal width.
- $f(x) = x^2$ between $x = 0$ and $x = 1$.
 - $f(x) = x^3$ between $x = 0$ and $x = 1$.
 - $f(x) = 1/x$ between $x = 1$ and $x = 5$.
 - $f(x) = 4 - x^2$ between $x = -2$ and $x = 2$.

Using rectangles whose height is given by the value of the function at the midpoint of the rectangle's base (*the midpoint rule*) estimate the area under the graphs of the following functions, using first two and then four rectangles.

- $f(x) = x^2$ between $x = 0$ and $x = 1$.
- $f(x) = x^3$ between $x = 0$ and $x = 1$.
- $f(x) = 1/x$ between $x = 1$ and $x = 5$.
- $f(x) = 4 - x^2$ between $x = -2$ and $x = 2$.

Distance

- Distance traveled** The accompanying table shows the velocity of a model train engine moving along a track for 10 sec. Estimate the distance traveled by the engine using 10 subintervals of length 1 with

- left-endpoint values.
- right-endpoint values.

Time (sec)	Velocity (in./sec)	Time (sec)	Velocity (in./sec)
0	0	6	11
1	12	7	6
2	22	8	2
3	10	9	6
4	5	10	0
5	13		

- Distance traveled upstream** You are sitting on the bank of a tidal river watching the incoming tide carry a bottle upstream. You record the velocity of the flow every 5 minutes for an hour, with the results shown in the accompanying table. About how far upstream did the bottle travel during that hour? Find an estimate using 12 subintervals of length 5 with

- left-endpoint values.
- right-endpoint values.

Time (min)	Velocity (m/sec)	Time (min)	Velocity (m/sec)
0	1	35	1.2
5	1.2	40	1.0
10	1.7	45	1.8
15	2.0	50	1.5
20	1.8	55	1.2
25	1.6	60	0
30	1.4		

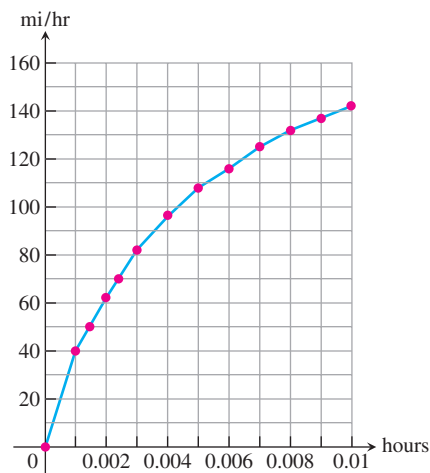
11. Length of a road You and a companion are about to drive a twisty stretch of dirt road in a car whose speedometer works but whose odometer (mileage counter) is broken. To find out how long this particular stretch of road is, you record the car's velocity at 10-sec intervals, with the results shown in the accompanying table. Estimate the length of the road using

- left-endpoint values.
- right-endpoint values.

Time (sec)	Velocity (converted to ft/sec) (30 mi/h = 44 ft/sec)	Time (sec)	Velocity (converted to ft/sec) (30 mi/h = 44 ft/sec)
0	0	70	15
10	44	80	22
20	15	90	35
30	35	100	44
40	30	110	30
50	44	120	35
60	35		

12. Distance from velocity data The accompanying table gives data for the velocity of a vintage sports car accelerating from 0 to 142 mi/h in 36 sec (10 thousandths of an hour).

Time (h)	Velocity (mi/h)	Time (h)	Velocity (mi/h)
0.0	0	0.006	116
0.001	40	0.007	125
0.002	62	0.008	132
0.003	82	0.009	137
0.004	96	0.010	142
0.005	108		



- Use rectangles to estimate how far the car traveled during the 36 sec it took to reach 142 mi/h.
- Roughly how many seconds did it take the car to reach the halfway point? About how fast was the car going then?

Velocity and Distance

13. Free fall with air resistance An object is dropped straight down from a helicopter. The object falls faster and faster but its acceleration (rate of change of its velocity) decreases over time because of air resistance. The acceleration is measured in ft/sec^2 and recorded every second after the drop for 5 sec, as shown:

t	0	1	2	3	4	5
a	32.00	19.41	11.77	7.14	4.33	2.63

- Find an upper estimate for the speed when $t = 5$.
- Find a lower estimate for the speed when $t = 5$.
- Find an upper estimate for the distance fallen when $t = 3$.

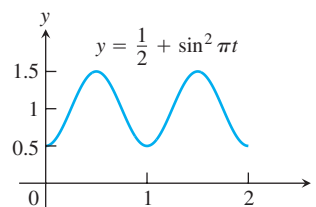
14. Distance traveled by a projectile An object is shot straight upward from sea level with an initial velocity of 400 ft/sec.

- Assuming that gravity is the only force acting on the object, give an upper estimate for its velocity after 5 sec have elapsed. Use $g = 32 \text{ ft/sec}^2$ for the gravitational acceleration.
- Find a lower estimate for the height attained after 5 sec.

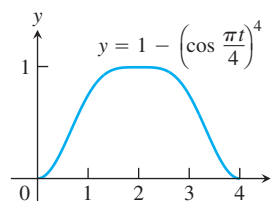
Average Value of a Function

In Exercises 15–18, use a finite sum to estimate the average value of f on the given interval by partitioning the interval into four subintervals of equal length and evaluating f at the subinterval midpoints.

- $f(x) = x^3$ on $[0, 2]$
- $f(x) = 1/x$ on $[1, 9]$
- $f(t) = (1/2) + \sin^2 \pi t$ on $[0, 2]$



- $f(t) = 1 - \left(\cos \frac{\pi t}{4}\right)^4$ on $[0, 4]$



Pollution Control

- 19. Water pollution** Oil is leaking out of a tanker damaged at sea. The damage to the tanker is worsening as evidenced by the increased leakage each hour, recorded in the following table.

Time (h)	0	1	2	3	4
Leakage (gal/h)	50	70	97	136	190

Time (h)	5	6	7	8
Leakage (gal/h)	265	369	516	720

- Give an upper and a lower estimate of the total quantity of oil that has escaped after 5 hours.
 - Repeat part (a) for the quantity of oil that has escaped after 8 hours.
 - The tanker continues to leak 720 gal/h after the first 8 hours. If the tanker originally contained 25,000 gal of oil, approximately how many more hours will elapse in the worst case before all the oil has spilled? In the best case?
- 20. Air pollution** A power plant generates electricity by burning oil. Pollutants produced as a result of the burning process are removed by scrubbers in the smokestacks. Over time, the scrubbers become less efficient and eventually they must be replaced when the amount of pollution released exceeds government standards. Measurements are taken at the end of each month determining the rate at which pollutants are released into the atmosphere, recorded as follows.

Month	Jan	Feb	Mar	Apr	May	Jun
Pollutant Release rate (tons/day)	0.20	0.25	0.27	0.34	0.45	0.52

Month	Jul	Aug	Sep	Oct	Nov	Dec
Pollutant Release rate (tons/day)	0.63	0.70	0.81	0.85	0.89	0.95

- Assuming a 30-day month and that new scrubbers allow only 0.05 ton/day released, give an upper estimate of the total tonnage of pollutants released by the end of June. What is a lower estimate?
- In the best case, approximately when will a total of 125 tons of pollutants have been released into the atmosphere?

Area of a Circle

- 21.** Inscribe a regular n -sided polygon inside a circle of radius 1 and compute the area of the polygon for the following values of n :
- 4 (square)
 - 8 (octagon)
 - 16
 - Compare the areas in parts (a), (b), and (c) with the area of the circle.
- 22.** (*Continuation of Exercise 21*)
- Inscribe a regular n -sided polygon inside a circle of radius 1 and compute the area of one of the n congruent triangles formed by drawing radii to the vertices of the polygon.
 - Compute the limit of the area of the inscribed polygon as $n \rightarrow \infty$.
 - Repeat the computations in parts (a) and (b) for a circle of radius r .

COMPUTER EXPLORATIONS

In Exercises 23–26, use a CAS to perform the following steps.

- Plot the functions over the given interval.
 - Subdivide the interval into $n = 100, 200$, and 1000 subintervals of equal length and evaluate the function at the midpoint of each subinterval.
 - Compute the average value of the function values generated in part (b).
 - Solve the equation $f(x) = (\text{average value})$ for x using the average value calculated in part (c) for the $n = 1000$ partitioning.
- 23.** $f(x) = \sin x$ on $[0, \pi]$ **24.** $f(x) = \sin^2 x$ on $[0, \pi]$
- 25.** $f(x) = x \sin \frac{1}{x}$ on $\left[\frac{\pi}{4}, \pi\right]$
- 26.** $f(x) = x \sin^2 \frac{1}{x}$ on $\left[\frac{\pi}{4}, \pi\right]$

5.2

Sigma Notation and Limits of Finite Sums

In estimating with finite sums in Section 5.1, we often encountered sums with many terms (up to 1000 in Table 5.1, for instance). In this section we introduce a notation to write sums with a large number of terms. After describing the notation and stating several of its properties, we look at what happens to a finite sum approximation as the number of terms approaches infinity.

Finite Sums and Sigma Notation

Sigma notation enables us to write a sum with many terms in the compact form

$$\sum_{k=1}^n a_k = a_1 + a_2 + a_3 + \cdots + a_{n-1} + a_n.$$

The Greek letter Σ (capital sigma, corresponding to our letter S), stands for “sum.” The **index of summation** k tells us where the sum begins (at the number below the Σ symbol) and where it ends (at the number above Σ). Any letter can be used to denote the index, but the letters i, j , and k are customary.

The index k ends at $k = n$.

$$\sum_{k=1}^n a_k$$

The summation symbol
(Greek letter sigma)

a_k is a formula for the k th term.

The index k starts at $k = 1$.

Thus we can write

$$1^2 + 2^2 + 3^2 + 4^2 + 5^2 + 6^2 + 7^2 + 8^2 + 9^2 + 10^2 + 11^2 = \sum_{k=1}^{11} k^2,$$

and

$$f(1) + f(2) + f(3) + \cdots + f(100) = \sum_{i=1}^{100} f(i).$$

The sigma notation used on the right side of these equations is much more compact than the summation expressions on the left side.

EXAMPLE 1 Using Sigma Notation

The sum in sigma notation	The sum written out, one term for each value of k	The value of the sum
$\sum_{k=1}^5 k$	$1 + 2 + 3 + 4 + 5$	15
$\sum_{k=1}^3 (-1)^k k$	$(-1)^1(1) + (-1)^2(2) + (-1)^3(3)$	$-1 + 2 - 3 = -2$
$\sum_{k=1}^2 \frac{k}{k+1}$	$\frac{1}{1+1} + \frac{2}{2+1}$	$\frac{1}{2} + \frac{2}{3} = \frac{7}{6}$
$\sum_{k=4}^5 \frac{k^2}{k-1}$	$\frac{4^2}{4-1} + \frac{5^2}{5-1}$	$\frac{16}{3} + \frac{25}{4} = \frac{139}{12}$

The lower limit of summation does not have to be 1; it can be any integer.

EXAMPLE 2 Using Different Index Starting Values

Express the sum $1 + 3 + 5 + 7 + 9$ in sigma notation.

Solution The formula generating the terms changes with the lower limit of summation, but the terms generated remain the same. It is often simplest to start with $k = 0$ or $k = 1$.

$$\text{Starting with } k = 0: \quad 1 + 3 + 5 + 7 + 9 = \sum_{k=0}^4 (2k + 1)$$

$$\text{Starting with } k = 1: \quad 1 + 3 + 5 + 7 + 9 = \sum_{k=1}^5 (2k - 1)$$

$$\text{Starting with } k = 2: \quad 1 + 3 + 5 + 7 + 9 = \sum_{k=2}^6 (2k - 3)$$

$$\text{Starting with } k = -3: \quad 1 + 3 + 5 + 7 + 9 = \sum_{k=-3}^1 (2k + 7) \quad \blacksquare$$

When we have a sum such as

$$\sum_{k=1}^3 (k + k^2)$$

we can rearrange its terms,

$$\begin{aligned} \sum_{k=1}^3 (k + k^2) &= (1 + 1^2) + (2 + 2^2) + (3 + 3^2) \\ &= (1 + 2 + 3) + (1^2 + 2^2 + 3^2) && \text{Regroup terms.} \\ &= \sum_{k=1}^3 k + \sum_{k=1}^3 k^2 \end{aligned}$$

This illustrates a general rule for finite sums:

$$\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k$$

Four such rules are given below. A proof that they are valid can be obtained using mathematical induction (see Appendix 1).

Algebra Rules for Finite Sums

1. *Sum Rule:* $\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k$
2. *Difference Rule:* $\sum_{k=1}^n (a_k - b_k) = \sum_{k=1}^n a_k - \sum_{k=1}^n b_k$
3. *Constant Multiple Rule:* $\sum_{k=1}^n c a_k = c \cdot \sum_{k=1}^n a_k$ (Any number c)
4. *Constant Value Rule:* $\sum_{k=1}^n c = n \cdot c$ (c is any constant value.)

EXAMPLE 3 Using the Finite Sum Algebra Rules

- (a) $\sum_{k=1}^n (3k - k^2) = 3 \sum_{k=1}^n k - \sum_{k=1}^n k^2$ Difference Rule
and Constant
Multiple Rule
- (b) $\sum_{k=1}^n (-a_k) = \sum_{k=1}^n (-1) \cdot a_k = -1 \cdot \sum_{k=1}^n a_k = -\sum_{k=1}^n a_k$ Constant
Multiple Rule
- (c) $\sum_{k=1}^3 (k + 4) = \sum_{k=1}^3 k + \sum_{k=1}^3 4$ Sum Rule
 $= (1 + 2 + 3) + (3 \cdot 4)$ Constant
Value Rule
 $= 6 + 12 = 18$
- (d) $\sum_{k=1}^n \frac{1}{n} = n \cdot \frac{1}{n} = 1$ Constant Value Rule
($1/n$ is constant) ■

HISTORICAL BIOGRAPHY

Carl Friedrich Gauss
(1777–1855)

Over the years people have discovered a variety of formulas for the values of finite sums. The most famous of these are the formula for the sum of the first n integers (Gauss may have discovered it at age 8) and the formulas for the sums of the squares and cubes of the first n integers.

EXAMPLE 4 The Sum of the First n Integers

Show that the sum of the first n integers is

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

Solution: The formula tells us that the sum of the first 4 integers is

$$\frac{(4)(5)}{2} = 10.$$

Addition verifies this prediction:

$$1 + 2 + 3 + 4 = 10.$$

To prove the formula in general, we write out the terms in the sum twice, once forward and once backward.

$$\begin{array}{ccccccccc} 1 & + & 2 & + & 3 & + & \cdots & + & n \\ n & + & (n-1) & + & (n-2) & + & \cdots & + & 1 \end{array}$$

If we add the two terms in the first column we get $1 + n = n + 1$. Similarly, if we add the two terms in the second column we get $2 + (n-1) = n + 1$. The two terms in any column sum to $n + 1$. When we add the n columns together we get n terms, each equal to $n + 1$, for a total of $n(n + 1)$. Since this is twice the desired quantity, the sum of the first n integers is $(n)(n + 1)/2$. ■

Formulas for the sums of the squares and cubes of the first n integers are proved using mathematical induction (see Appendix 1). We state them here.

The first n squares:	$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$
The first n cubes:	$\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2}\right)^2$

Limits of Finite Sums

The finite sum approximations we considered in Section 5.1 got more accurate as the number of terms increased and the subinterval widths (lengths) became thinner. The next example shows how to calculate a limiting value as the widths of the subintervals go to zero and their number grows to infinity.

EXAMPLE 5 The Limit of Finite Approximations to an Area

Find the limiting value of lower sum approximations to the area of the region R below the graph of $y = 1 - x^2$ and above the interval $[0, 1]$ on the x -axis using equal width rectangles whose widths approach zero and whose number approaches infinity. (See Figure 5.4a.)

Solution We compute a lower sum approximation using n rectangles of equal width $\Delta x = (1 - 0)/n$, and then we see what happens as $n \rightarrow \infty$. We start by subdividing $[0, 1]$ into n equal width subintervals

$$\left[0, \frac{1}{n}\right], \left[\frac{1}{n}, \frac{2}{n}\right], \dots, \left[\frac{n-1}{n}, 1\right].$$

Each subinterval has width $1/n$. The function $1 - x^2$ is decreasing on $[0, 1]$, and its smallest value in a subinterval occurs at the subinterval's right endpoint. So a lower sum is constructed with rectangles whose height over the subinterval $[(k-1)/n, k/n]$ is $f(k/n) = 1 - (k/n)^2$, giving the sum

$$f\left(\frac{1}{n}\right)\left(\frac{1}{n}\right) + f\left(\frac{2}{n}\right)\left(\frac{1}{n}\right) + \cdots + f\left(\frac{k}{n}\right)\left(\frac{1}{n}\right) + \cdots + f\left(\frac{n}{n}\right)\left(\frac{1}{n}\right).$$

We write this in sigma notation and simplify,

$$\begin{aligned} \sum_{k=1}^n f\left(\frac{k}{n}\right)\left(\frac{1}{n}\right) &= \sum_{k=1}^n \left(1 - \left(\frac{k}{n}\right)^2\right)\left(\frac{1}{n}\right) \\ &= \sum_{k=1}^n \left(\frac{1}{n} - \frac{k^2}{n^3}\right) \\ &= \sum_{k=1}^n \frac{1}{n} - \sum_{k=1}^n \frac{k^2}{n^3} && \text{Difference Rule} \\ &= n \cdot \frac{1}{n} - \frac{1}{n^3} \sum_{k=1}^n k^2 && \text{Constant Value and Constant Multiple Rules} \\ &= 1 - \left(\frac{1}{n^3}\right) \frac{(n)(n+1)(2n+1)}{6} && \text{Sum of the First } n \text{ Squares} \\ &= 1 - \frac{2n^3 + 3n^2 + n}{6n^3}. && \text{Numerator expanded} \end{aligned}$$

We have obtained an expression for the lower sum that holds for any n . Taking the limit of this expression as $n \rightarrow \infty$, we see that the lower sums converge as the number of subintervals increases and the subinterval widths approach zero:

$$\lim_{n \rightarrow \infty} \left(1 - \frac{2n^3 + 3n^2 + n}{6n^3}\right) = 1 - \frac{2}{6} = \frac{2}{3}.$$

The lower sum approximations converge to $2/3$. A similar calculation shows that the upper sum approximations also converge to $2/3$ (Exercise 35). Any finite sum approximation, in the sense of our summary at the end of Section 5.1, also converges to the same value

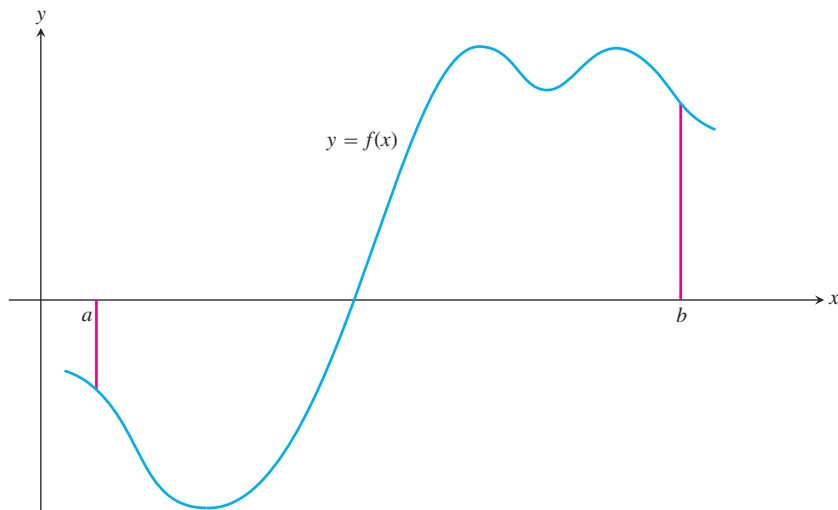


FIGURE 5.8 A typical continuous function $y = f(x)$ over a closed interval $[a, b]$.

$2/3$. This is because it is possible to show that any finite sum approximation is trapped between the lower and upper sum approximations. For this reason we are led to *define* the area of the region R as this limiting value. In Section 5.3 we study the limits of such finite approximations in their more general setting. ■

Riemann Sums

HISTORICAL BIOGRAPHY

Georg Friedrich
Bernhard Riemann
(1826–1866)

The theory of limits of finite approximations was made precise by the German mathematician Bernhard Riemann. We now introduce the notion of a *Riemann sum*, which underlies the theory of the definite integral studied in the next section.

We begin with an arbitrary function f defined on a closed interval $[a, b]$. Like the function pictured in Figure 5.8, f may have negative as well as positive values. We subdivide the interval $[a, b]$ into subintervals, not necessarily of equal widths (or lengths), and form sums in the same way as for the finite approximations in Section 5.1. To do so, we choose $n - 1$ points $\{x_1, x_2, x_3, \dots, x_{n-1}\}$ between a and b and satisfying

$$a < x_1 < x_2 < \cdots < x_{n-1} < b.$$

To make the notation consistent, we denote a by x_0 and b by x_n , so that

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b.$$

The set

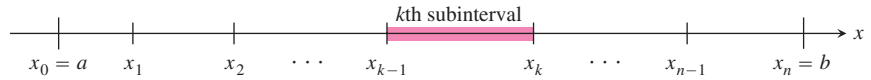
$$P = \{x_0, x_1, x_2, \dots, x_{n-1}, x_n\}$$

is called a **partition** of $[a, b]$.

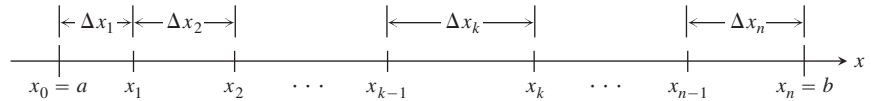
The partition P divides $[a, b]$ into n closed subintervals

$$[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n].$$

The first of these subintervals is $[x_0, x_1]$, the second is $[x_1, x_2]$, and the k th subinterval of P is $[x_{k-1}, x_k]$, for k an integer between 1 and n .



The width of the first subinterval $[x_0, x_1]$ is denoted Δx_1 , the width of the second $[x_1, x_2]$ is denoted Δx_2 , and the width of the k th subinterval is $\Delta x_k = x_k - x_{k-1}$. If all n subintervals have equal width, then the common width Δx is equal to $(b - a)/n$.



In each subinterval we select some point. The point chosen in the k th subinterval $[x_{k-1}, x_k]$ is called c_k . Then on each subinterval we stand a vertical rectangle that stretches from the x -axis to touch the curve at $(c_k, f(c_k))$. These rectangles can be above or below the x -axis, depending on whether $f(c_k)$ is positive or negative, or on it if $f(c_k) = 0$ (Figure 5.9).

On each subinterval we form the product $f(c_k) \cdot \Delta x_k$. This product is positive, negative or zero, depending on the sign of $f(c_k)$. When $f(c_k) > 0$, the product $f(c_k) \cdot \Delta x_k$ is the area of a rectangle with height $f(c_k)$ and width Δx_k . When $f(c_k) < 0$, the product $f(c_k) \cdot \Delta x_k$ is a negative number, the negative of the area of a rectangle of width Δx_k that drops from the x -axis to the negative number $f(c_k)$.

Finally we sum all these products to get

$$S_P = \sum_{k=1}^n f(c_k) \Delta x_k.$$

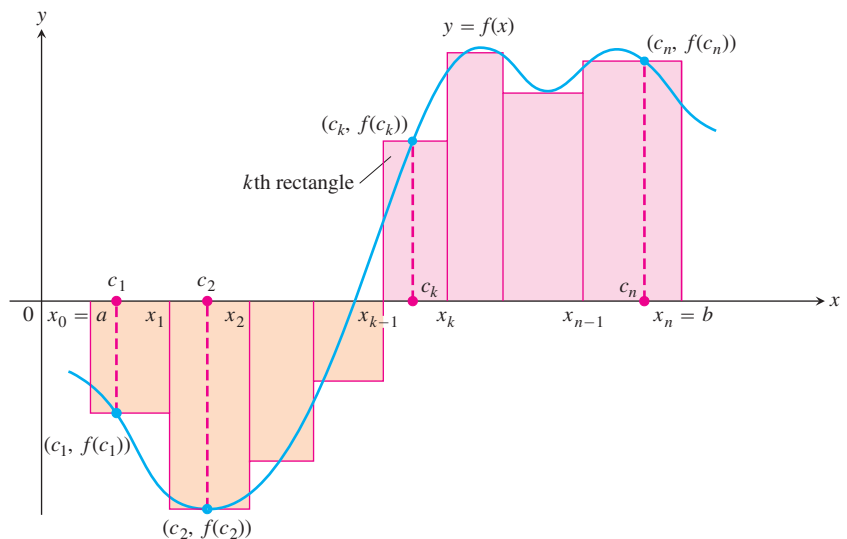


FIGURE 5.9 The rectangles approximate the region between the graph of the function $y = f(x)$ and the x -axis.

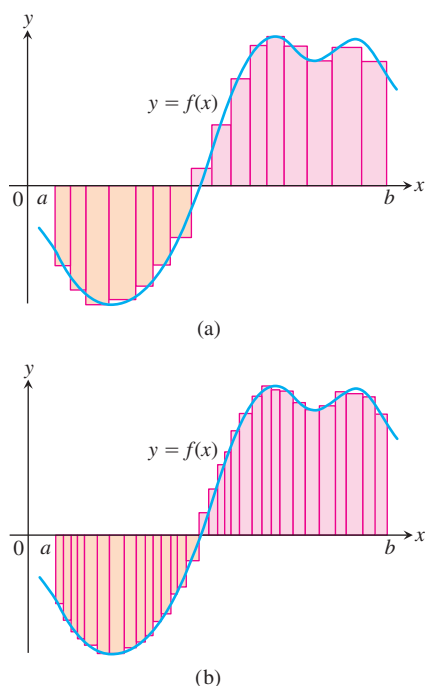


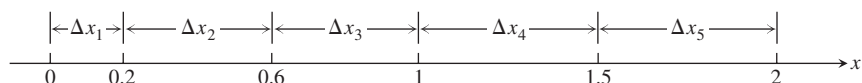
FIGURE 5.10 The curve of Figure 5.9 with rectangles from finer partitions of $[a, b]$. Finer partitions create collections of rectangles with thinner bases that approximate the region between the graph of f and the x -axis with increasing accuracy.

The sum S_P is called a **Riemann sum for f on the interval $[a, b]$** . There are many such sums, depending on the partition P we choose, and the choices of the points c_k in the subintervals.

In Example 5, where the subintervals all had equal widths $\Delta x = 1/n$, we could make them thinner by simply increasing their number n . When a partition has subintervals of varying widths, we can ensure they are all thin by controlling the width of a widest (longest) subinterval. We define the **norm** of a partition P , written $\|P\|$, to be the largest of all the subinterval widths. If $\|P\|$ is a small number, then all of the subintervals in the partition P have a small width. Let's look at an example of these ideas.

EXAMPLE 6 Partitioning a Closed Interval

The set $P = \{0, 0.2, 0.6, 1, 1.5, 2\}$ is a partition of $[0, 2]$. There are five subintervals of P : $[0, 0.2]$, $[0.2, 0.6]$, $[0.6, 1]$, $[1, 1.5]$, and $[1.5, 2]$:



The lengths of the subintervals are $\Delta x_1 = 0.2$, $\Delta x_2 = 0.4$, $\Delta x_3 = 0.4$, $\Delta x_4 = 0.5$, and $\Delta x_5 = 0.5$. The longest subinterval length is 0.5, so the norm of the partition is $\|P\| = 0.5$. In this example, there are two subintervals of this length. ■

Any Riemann sum associated with a partition of a closed interval $[a, b]$ defines rectangles that approximate the region between the graph of a continuous function f and the x -axis. Partitions with norm approaching zero lead to collections of rectangles that approximate this region with increasing accuracy, as suggested by Figure 5.10. We will see in the next section that if the function f is continuous over the closed interval $[a, b]$, then no matter how we choose the partition P and the points c_k in its subintervals to construct a Riemann sum, a single limiting value is approached as the subinterval widths, controlled by the norm of the partition, approach zero.

EXERCISES 5.2

Sigma Notation

Write the sums in Exercises 1–6 without sigma notation. Then evaluate them.

1. $\sum_{k=1}^2 \frac{6k}{k+1}$

2. $\sum_{k=1}^3 \frac{k-1}{k}$

3. $\sum_{k=1}^4 \cos k\pi$

4. $\sum_{k=1}^5 \sin k\pi$

5. $\sum_{k=1}^3 (-1)^{k+1} \sin \frac{\pi}{k}$

6. $\sum_{k=1}^4 (-1)^k \cos k\pi$

7. Which of the following express $1 + 2 + 4 + 8 + 16 + 32$ in sigma notation?

a. $\sum_{k=1}^6 2^{k-1}$

b. $\sum_{k=0}^5 2^k$

c. $\sum_{k=-1}^4 2^{k+1}$

8. Which of the following express $1 - 2 + 4 - 8 + 16 - 32$ in sigma notation?

a. $\sum_{k=1}^6 (-2)^{k-1}$

b. $\sum_{k=0}^5 (-1)^k 2^k$

c. $\sum_{k=-2}^3 (-1)^{k+1} 2^{k+2}$

9. Which formula is not equivalent to the other two?

a. $\sum_{k=2}^4 \frac{(-1)^{k-1}}{k-1}$

b. $\sum_{k=0}^2 \frac{(-1)^k}{k+1}$

c. $\sum_{k=-1}^1 \frac{(-1)^k}{k+2}$

10. Which formula is not equivalent to the other two?

a. $\sum_{k=1}^4 (k-1)^2$

b. $\sum_{k=-1}^3 (k+1)^2$

c. $\sum_{k=-3}^{-1} k^2$

Express the sums in Exercises 11–16 in sigma notation. The form of your answer will depend on your choice of the lower limit of summation.

11. $1 + 2 + 3 + 4 + 5 + 6$ 12. $1 + 4 + 9 + 16$

13. $\frac{1}{2} + \frac{1}{4} + \frac{1}{8} + \frac{1}{16}$ 14. $2 + 4 + 6 + 8 + 10$

15. $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5}$ 16. $-\frac{1}{5} + \frac{2}{5} - \frac{3}{5} + \frac{4}{5} - \frac{5}{5}$

Values of Finite Sums

17. Suppose that $\sum_{k=1}^n a_k = -5$ and $\sum_{k=1}^n b_k = 6$. Find the values of

a. $\sum_{k=1}^n 3a_k$ b. $\sum_{k=1}^n \frac{b_k}{6}$ c. $\sum_{k=1}^n (a_k + b_k)$

d. $\sum_{k=1}^n (a_k - b_k)$ e. $\sum_{k=1}^n (b_k - 2a_k)$

18. Suppose that $\sum_{k=1}^n a_k = 0$ and $\sum_{k=1}^n b_k = 1$. Find the values of

a. $\sum_{k=1}^n 8a_k$ b. $\sum_{k=1}^n 250b_k$

c. $\sum_{k=1}^n (a_k + 1)$ d. $\sum_{k=1}^n (b_k - 1)$

Evaluate the sums in Exercises 19–28.

19. a. $\sum_{k=1}^{10} k$ b. $\sum_{k=1}^{10} k^2$ c. $\sum_{k=1}^{10} k^3$

20. a. $\sum_{k=1}^{13} k$ b. $\sum_{k=1}^{13} k^2$ c. $\sum_{k=1}^{13} k^3$

21. $\sum_{k=1}^7 (-2k)$ 22. $\sum_{k=1}^5 \frac{\pi k}{15}$

23. $\sum_{k=1}^6 (3 - k^2)$ 24. $\sum_{k=1}^6 (k^2 - 5)$

25. $\sum_{k=1}^5 k(3k + 5)$

26. $\sum_{k=1}^7 k(2k + 1)$

27. $\sum_{k=1}^5 \frac{k^3}{225} + \left(\sum_{k=1}^5 k \right)^3$

28. $\left(\sum_{k=1}^7 k \right)^2 - \sum_{k=1}^7 \frac{k^3}{4}$

Rectangles for Riemann Sums

In Exercises 29–32, graph each function $f(x)$ over the given interval. Partition the interval into four subintervals of equal length. Then add to your sketch the rectangles associated with the Riemann sum $\sum_{k=1}^4 f(c_k) \Delta x_k$, given that c_k is the (a) left-hand endpoint, (b) right-hand endpoint, (c) midpoint of the k th subinterval. (Make a separate sketch for each set of rectangles.)

29. $f(x) = x^2 - 1$, $[0, 2]$

30. $f(x) = -x^2$, $[0, 1]$

31. $f(x) = \sin x$, $[-\pi, \pi]$

32. $f(x) = \sin x + 1$, $[-\pi, \pi]$

33. Find the norm of the partition $P = \{0, 1.2, 1.5, 2.3, 2.6, 3\}$.

34. Find the norm of the partition $P = \{-2, -1.6, -0.5, 0, 0.8, 1\}$.

Limits of Upper Sums

For the functions in Exercises 35–40 find a formula for the upper sum obtained by dividing the interval $[a, b]$ into n equal subintervals. Then take a limit of these sums as $n \rightarrow \infty$ to calculate the area under the curve over $[a, b]$.

35. $f(x) = 1 - x^2$ over the interval $[0, 1]$.

36. $f(x) = 2x$ over the interval $[0, 3]$.

37. $f(x) = x^2 + 1$ over the interval $[0, 3]$.

38. $f(x) = 3x^2$ over the interval $[0, 1]$.

39. $f(x) = x + x^2$ over the interval $[0, 1]$.

40. $f(x) = 3x + 2x^2$ over the interval $[0, 1]$.

5.3

The Definite Integral

In Section 5.2 we investigated the limit of a finite sum for a function defined over a closed interval $[a, b]$ using n subintervals of equal width (or length), $(b - a)/n$. In this section we consider the limit of more general Riemann sums as the norm of the partitions of $[a, b]$ approaches zero. For general Riemann sums the subintervals of the partitions need not have equal widths. The limiting process then leads to the definition of the *definite integral* of a function over a closed interval $[a, b]$.

Limits of Riemann Sums

The definition of the definite integral is based on the idea that for certain functions, as the norm of the partitions of $[a, b]$ approaches zero, the values of the corresponding Riemann

sums approach a limiting value I . What we mean by this converging idea is that a Riemann sum will be close to the number I provided that the norm of its partition is sufficiently small (so that all of its subintervals have thin enough widths). We introduce the symbol ϵ as a small positive number that specifies how close to I the Riemann sum must be, and the symbol δ as a second small positive number that specifies how small the norm of a partition must be in order for that to happen. Here is a precise formulation.

DEFINITION The Definite Integral as a Limit of Riemann Sums

Let $f(x)$ be a function defined on a closed interval $[a, b]$. We say that a number I is the **definite integral of f over $[a, b]$** and that I is the limit of the Riemann sums $\sum_{k=1}^n f(c_k) \Delta x_k$ if the following condition is satisfied:

Given any number $\epsilon > 0$ there is a corresponding number $\delta > 0$ such that for every partition $P = \{x_0, x_1, \dots, x_n\}$ of $[a, b]$ with $\|P\| < \delta$ and any choice of c_k in $[x_{k-1}, x_k]$, we have

$$\left| \sum_{k=1}^n f(c_k) \Delta x_k - I \right| < \epsilon.$$

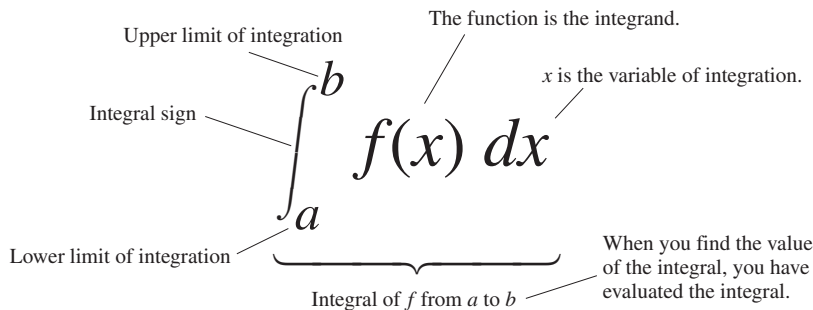
Leibniz introduced a notation for the definite integral that captures its construction as a limit of Riemann sums. He envisioned the finite sums $\sum_{k=1}^n f(c_k) \Delta x_k$ becoming an infinite sum of function values $f(x)$ multiplied by “infinitesimal” subinterval widths dx . The sum symbol $\sum$ is replaced in the limit by the integral symbol $\int$, whose origin is in the letter “S.” The function values $f(c_k)$ are replaced by a continuous selection of function values $f(x)$. The subinterval widths Δx_k become the differential dx . It is as if we are summing all products of the form $f(x) \cdot dx$ as x goes from a to b . While this notation captures the process of constructing an integral, it is Riemann’s definition that gives a precise meaning to the definite integral.

Notation and Existence of the Definite Integral

The symbol for the number I in the definition of the definite integral is

$$\int_a^b f(x) dx$$

which is read as “the integral from a to b of f of x dee x ” or sometimes as “the integral from a to b of f of x with respect to x .” The component parts in the integral symbol also have names:



When the definition is satisfied, we say the Riemann sums of f on $[a, b]$ **converge** to the definite integral $I = \int_a^b f(x) dx$ and that f is **integrable** over $[a, b]$. We have many choices for a partition P with norm going to zero, and many choices of points c_k for each partition. The definite integral exists when we always get the same limit I , no matter what choices are made. When the limit exists we write it as the definite integral

$$\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k = I = \int_a^b f(x) dx.$$

When each partition has n equal subintervals, each of width $\Delta x = (b - a)/n$, we will also write

$$\lim_{n \rightarrow \infty} \sum_{k=1}^n f(c_k) \Delta x = I = \int_a^b f(x) dx.$$

The limit is always taken as the norm of the partitions approaches zero and the number of subintervals goes to infinity.

The value of the definite integral of a function over any particular interval depends on the function, not on the letter we choose to represent its independent variable. If we decide to use t or u instead of x , we simply write the integral as

$$\int_a^b f(t) dt \quad \text{or} \quad \int_a^b f(u) du \quad \text{instead of} \quad \int_a^b f(x) dx.$$

No matter how we write the integral, it is still the same number, defined as a limit of Riemann sums. Since it does not matter what letter we use, the variable of integration is called a **dummy variable**.

Since there are so many choices to be made in taking a limit of Riemann sums, it might seem difficult to show that such a limit exists. It turns out, however, that no matter what choices are made, the Riemann sums associated with a *continuous* function converge to the same limit.

THEOREM 1 The Existence of Definite Integrals

A continuous function is integrable. That is, if a function f is continuous on an interval $[a, b]$, then its definite integral over $[a, b]$ exists.

By the Extreme Value Theorem (Theorem 1, Section 4.1), when f is continuous we can choose c_k so that $f(c_k)$ gives the maximum value of f on $[x_{k-1}, x_k]$, giving an **upper sum**. We can choose c_k to give the minimum value of f on $[x_{k-1}, x_k]$, giving a **lower sum**. We can pick c_k to be the midpoint of $[x_{k-1}, x_k]$, the rightmost point x_k , or a random point. We can take the partitions of equal or varying widths. In each case we get the same limit for $\sum_{k=1}^n f(c_k) \Delta x_k$ as $\|P\| \rightarrow 0$. The idea behind Theorem 1 is that a Riemann sum associated with a partition is no more than the upper sum of that partition and no less than the lower sum. The upper and lower sums converge to the same value when $\|P\| \rightarrow 0$. All other Riemann sums lie between the upper and lower sums and have the same limit. A proof of Theorem 1 involves a careful analysis of functions, partitions, and limits along this line of thinking and is left to a more advanced text. An indication of this proof is given in Exercises 80 and 81.

Theorem 1 says nothing about how to *calculate* definite integrals. A method of calculation will be developed in Section 5.4, through a connection to the process of taking anti-derivatives.

Integrable and Nonintegrable Functions

Theorem 1 tells us that functions continuous over the interval $[a, b]$ are integrable there. Functions that are not continuous may or may not be integrable. Discontinuous functions that are integrable include those that are increasing on $[a, b]$ (Exercise 77), and the *piecewise-continuous functions* defined in the Additional Exercises at the end of this chapter. (The latter are continuous except at a finite number of points in $[a, b]$.) For integrability to fail, a function needs to be sufficiently discontinuous so that the region between its graph and the x -axis cannot be approximated well by increasingly thin rectangles. Here is an example of a function that is not integrable.

EXAMPLE 1 A Nonintegrable Function on $[0, 1]$

The function

$$f(x) = \begin{cases} 1, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$$

has no Riemann integral over $[0, 1]$. Underlying this is the fact that between any two numbers there is both a rational number and an irrational number. Thus the function jumps up and down too erratically over $[0, 1]$ to allow the region beneath its graph and above the x -axis to be approximated by rectangles, no matter how thin they are. We show, in fact, that upper sum approximations and lower sum approximations converge to different limiting values.

If we pick a partition P of $[0, 1]$ and choose c_k to be the maximum value for f on $[x_{k-1}, x_k]$ then the corresponding Riemann sum is

$$U = \sum_{k=1}^n f(c_k) \Delta x_k = \sum_{k=1}^n (1) \Delta x_k = 1,$$

since each subinterval $[x_{k-1}, x_k]$ contains a rational number where $f(c_k) = 1$. Note that the lengths of the intervals in the partition sum to 1, $\sum_{k=1}^n \Delta x_k = 1$. So each such Riemann sum equals 1, and a limit of Riemann sums using these choices equals 1.

On the other hand, if we pick c_k to be the minimum value for f on $[x_{k-1}, x_k]$, then the Riemann sum is

$$L = \sum_{k=1}^n f(c_k) \Delta x_k = \sum_{k=1}^n (0) \Delta x_k = 0,$$

since each subinterval $[x_{k-1}, x_k]$ contains an irrational number c_k where $f(c_k) = 0$. The limit of Riemann sums using these choices equals zero. Since the limit depends on the choices of c_k , the function f is not integrable. ■

Properties of Definite Integrals

In defining $\int_a^b f(x) dx$ as a limit of sums $\sum_{k=1}^n f(c_k) \Delta x_k$, we moved from left to right across the interval $[a, b]$. What would happen if we instead move right to left, starting with $x_0 = b$ and ending at $x_n = a$. Each Δx_k in the Riemann sum would change its sign, with $x_k - x_{k-1}$ now negative instead of positive. With the same choices of c_k in each subinterval, the sign of any Riemann sum would change, as would the sign of the limit, the integral

$\int_b^a f(x) dx$. Since we have not previously given a meaning to integrating backward, we are led to define

$$\int_b^a f(x) dx = -\int_a^b f(x) dx.$$

Another extension of the integral is to an interval of zero width, when $a = b$. Since $f(c_k) \Delta x_k$ is zero when the interval width $\Delta x_k = 0$, we define

$$\int_a^a f(x) dx = 0.$$

Theorem 2 states seven properties of integrals, given as rules that they satisfy, including the two above. These rules become very useful in the process of computing integrals. We will refer to them repeatedly to simplify our calculations.

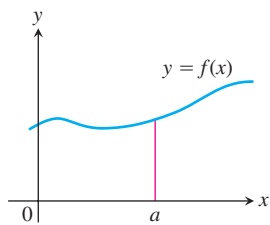
Rules 2 through 7 have geometric interpretations, shown in Figure 5.11. The graphs in these figures are of positive functions, but the rules apply to general integrable functions.

THEOREM 2

When f and g are integrable, the definite integral satisfies Rules 1 to 7 in Table 5.3.

TABLE 5.3 Rules satisfied by definite integrals

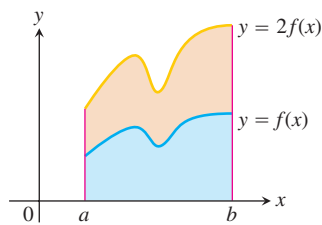
1. <i>Order of Integration:</i>	$\int_b^a f(x) dx = -\int_a^b f(x) dx$	A Definition
2. <i>Zero Width Interval:</i>	$\int_a^a f(x) dx = 0$	Also a Definition
3. <i>Constant Multiple:</i>	$\int_a^b kf(x) dx = k \int_a^b f(x) dx$	Any Number k
	$\int_a^b -f(x) dx = -\int_a^b f(x) dx$	$k = -1$
4. <i>Sum and Difference:</i>	$\int_a^b (f(x) \pm g(x)) dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$	
5. <i>Additivity:</i>	$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$	
6. <i>Max-Min Inequality:</i>	If f has maximum value $\max f$ and minimum value $\min f$ on $[a, b]$, then	
	$\min f \cdot (b - a) \leq \int_a^b f(x) dx \leq \max f \cdot (b - a).$	
7. <i>Domination:</i>	$f(x) \geq g(x) \text{ on } [a, b] \Rightarrow \int_a^b f(x) dx \geq \int_a^b g(x) dx$	
	$f(x) \geq 0 \text{ on } [a, b] \Rightarrow \int_a^b f(x) dx \geq 0$	(Special Case)



(a) *Zero Width Interval:*

$$\int_a^a f(x) dx = 0.$$

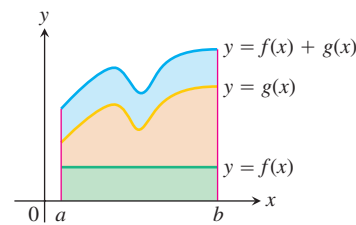
(The area over a point is 0.)



(b) *Constant Multiple:*

$$\int_a^b kf(x) dx = k \int_a^b f(x) dx.$$

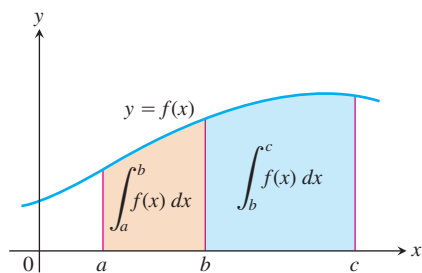
(Shown for $k = 2$.)



(c) *Sum:*

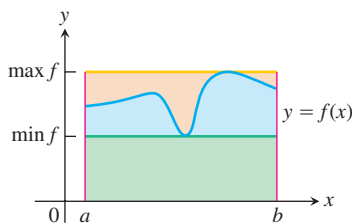
$$\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$$

(Areas add)



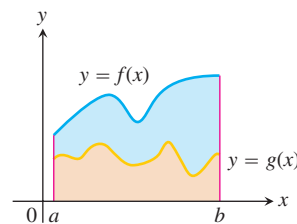
(d) *Additivity for definite integrals:*

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$$



(e) *Max-Min Inequality:*

$$\begin{aligned} \min f \cdot (b - a) &\leq \int_a^b f(x) dx \\ &\leq \max f \cdot (b - a) \end{aligned}$$



(f) *Domination:*

$$\begin{aligned} f(x) &\geq g(x) \text{ on } [a, b] \\ \Rightarrow \int_a^b f(x) dx &\geq \int_a^b g(x) dx \end{aligned}$$

FIGURE 5.11

While Rules 1 and 2 are definitions, Rules 3 to 7 of Table 5.3 must be proved. The proofs are based on the definition of the definite integral as a limit of Riemann sums. The following is a proof of one of these rules. Similar proofs can be given to verify the other properties in Table 5.3.

Proof of Rule 6 Rule 6 says that the integral of f over $[a, b]$ is never smaller than the minimum value of f times the length of the interval and never larger than the maximum value of f times the length of the interval. The reason is that for every partition of $[a, b]$ and for every choice of the points c_k ,

$$\begin{aligned} \min f \cdot (b - a) &= \min f \cdot \sum_{k=1}^n \Delta x_k && \sum_{k=1}^n \Delta x_k = b - a \\ &= \sum_{k=1}^n \min f \cdot \Delta x_k && \text{Constant Multiple Rule} \\ &\leq \sum_{k=1}^n f(c_k) \Delta x_k && \min f \leq f(c_k) \\ &\leq \sum_{k=1}^n \max f \cdot \Delta x_k && f(c_k) \leq \max f \\ &= \max f \cdot \sum_{k=1}^n \Delta x_k && \text{Constant Multiple Rule} \\ &= \max f \cdot (b - a). \end{aligned}$$

In short, all Riemann sums for f on $[a, b]$ satisfy the inequality

$$\min f \cdot (b - a) \leq \sum_{k=1}^n f(c_k) \Delta x_k \leq \max f \cdot (b - a).$$

Hence their limit, the integral, does too. ■

EXAMPLE 2 Using the Rules for Definite Integrals

Suppose that

$$\int_{-1}^1 f(x) dx = 5, \quad \int_1^4 f(x) dx = -2, \quad \int_{-1}^1 h(x) dx = 7.$$

Then

1. $\int_4^1 f(x) dx = -\int_1^4 f(x) dx = -(-2) = 2$ Rule 1
2. $\int_{-1}^1 [2f(x) + 3h(x)] dx = 2 \int_{-1}^1 f(x) dx + 3 \int_{-1}^1 h(x) dx$ Rules 3 and 4
 $= 2(5) + 3(7) = 31$
3. $\int_{-1}^4 f(x) dx = \int_{-1}^1 f(x) dx + \int_1^4 f(x) dx = 5 + (-2) = 3$ Rule 5 ■

EXAMPLE 3 Finding Bounds for an Integral

Show that the value of $\int_0^1 \sqrt{1 + \cos x} dx$ is less than $3/2$.

Solution The Max-Min Inequality for definite integrals (Rule 6) says that $\min f \cdot (b - a)$ is a *lower bound* for the value of $\int_a^b f(x) dx$ and that $\max f \cdot (b - a)$ is an *upper bound*. The maximum value of $\sqrt{1 + \cos x}$ on $[0, 1]$ is $\sqrt{1 + 1} = \sqrt{2}$, so

$$\int_0^1 \sqrt{1 + \cos x} dx \leq \sqrt{2} \cdot (1 - 0) = \sqrt{2}.$$

Since $\int_0^1 \sqrt{1 + \cos x} dx$ is bounded from above by $\sqrt{2}$ (which is 1.414 ...), the integral is less than $3/2$. ■

Area Under the Graph of a Nonnegative Function

We now make precise the notion of the area of a region with curved boundary, capturing the idea of approximating a region by increasingly many rectangles. The area under the graph of a nonnegative continuous function is defined to be a definite integral.

DEFINITION Area Under a Curve as a Definite Integral

If $y = f(x)$ is nonnegative and integrable over a closed interval $[a, b]$, then the **area under the curve $y = f(x)$ over $[a, b]$** is the integral of f from a to b ,

$$A = \int_a^b f(x) dx.$$

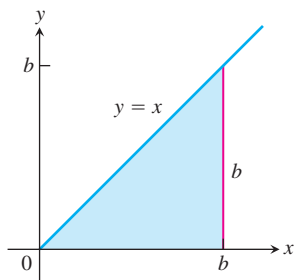


FIGURE 5.12 The region in Example 4 is a triangle.

For the first time we have a rigorous definition for the area of a region whose boundary is the graph of any continuous function. We now apply this to a simple example, the area under a straight line, where we can verify that our new definition agrees with our previous notion of area.

EXAMPLE 4 Area Under the Line $y = x$

Compute $\int_0^b x \, dx$ and find the area A under $y = x$ over the interval $[0, b]$, $b > 0$.

Solution The region of interest is a triangle (Figure 5.12). We compute the area in two ways.

- (a) To compute the definite integral as the limit of Riemann sums, we calculate $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n f(c_k) \Delta x_k$ for partitions whose norms go to zero. Theorem 1 tells us that it does not matter how we choose the partitions or the points c_k as long as the norms approach zero. All choices give the exact same limit. So we consider the partition P that subdivides the interval $[0, b]$ into n subintervals of equal width $\Delta x = (b - 0)/n = b/n$, and we choose c_k to be the right endpoint in each subinterval. The partition is $P = \left\{ 0, \frac{b}{n}, \frac{2b}{n}, \frac{3b}{n}, \dots, \frac{nb}{n} \right\}$ and $c_k = \frac{kb}{n}$. So

$$\begin{aligned} \sum_{k=1}^n f(c_k) \Delta x &= \sum_{k=1}^n \frac{kb}{n} \cdot \frac{b}{n} && f(c_k) = c_k \\ &= \sum_{k=1}^n \frac{kb^2}{n^2} \\ &= \frac{b^2}{n^2} \sum_{k=1}^n k && \text{Constant Multiple Rule} \\ &= \frac{b^2}{n^2} \cdot \frac{n(n+1)}{2} && \text{Sum of First } n \text{ Integers} \\ &= \frac{b^2}{2} \left(1 + \frac{1}{n} \right) \end{aligned}$$

As $n \rightarrow \infty$ and $\|P\| \rightarrow 0$, this last expression on the right has the limit $b^2/2$. Therefore,

$$\int_0^b x \, dx = \frac{b^2}{2}.$$

- (b) Since the area equals the definite integral for a nonnegative function, we can quickly derive the definite integral by using the formula for the area of a triangle having base length b and height $y = b$. The area is $A = (1/2) b \cdot b = b^2/2$. Again we have that $\int_0^b x \, dx = b^2/2$. ■

Example 4 can be generalized to integrate $f(x) = x$ over any closed interval $[a, b]$, $0 < a < b$.

$$\begin{aligned} \int_a^b x \, dx &= \int_a^0 x \, dx + \int_0^b x \, dx && \text{Rule 5} \\ &= -\int_0^a x \, dx + \int_0^b x \, dx && \text{Rule 1} \\ &= -\frac{a^2}{2} + \frac{b^2}{2}. && \text{Example 4} \end{aligned}$$

In conclusion, we have the following rule for integrating $f(x) = x$:

$$\int_a^b x \, dx = \frac{b^2}{2} - \frac{a^2}{2}, \quad a < b \quad (1)$$

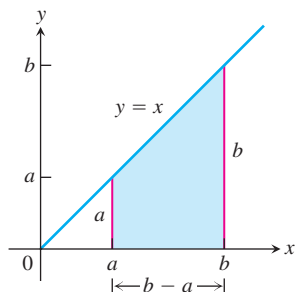


FIGURE 5.13 The area of this trapezoidal region is $A = (b^2 - a^2)/2$.

This computation gives the area of a trapezoid (Figure 5.13). Equation (1) remains valid when a and b are negative. When $a < b < 0$, the definite integral value $(b^2 - a^2)/2$ is a negative number, the negative of the area of a trapezoid dropping down to the line $y = x$ below the x -axis. When $a < 0$ and $b > 0$, Equation (1) is still valid and the definite integral gives the difference between two areas, the area under the graph and above $[0, b]$ minus the area below $[a, 0]$ and over the graph.

The following results can also be established using a Riemann sum calculation similar to that in Example 4 (Exercises 75 and 76).

$$\int_a^b c \, dx = c(b - a), \quad c \text{ any constant} \quad (2)$$

$$\int_a^b x^2 \, dx = \frac{b^3}{3} - \frac{a^3}{3}, \quad a < b \quad (3)$$

Average Value of a Continuous Function Revisited

In Section 5.1 we introduced informally the average value of a nonnegative continuous function f over an interval $[a, b]$, leading us to define this average as the area under the graph of $y = f(x)$ divided by $b - a$. In integral notation we write this as

$$\text{Average} = \frac{1}{b - a} \int_a^b f(x) \, dx.$$

We can use this formula to give a precise definition of the average value of any continuous (or integrable) function, whether positive, negative or both.

Alternately, we can use the following reasoning. We start with the idea from arithmetic that the average of n numbers is their sum divided by n . A continuous function f on $[a, b]$ may have infinitely many values, but we can still sample them in an orderly way. We divide $[a, b]$ into n subintervals of equal width $\Delta x = (b - a)/n$ and evaluate f at a point c_k in each (Figure 5.14). The average of the n sampled values is

$$\begin{aligned} \frac{f(c_1) + f(c_2) + \cdots + f(c_n)}{n} &= \frac{1}{n} \sum_{k=1}^n f(c_k) \\ &= \frac{\Delta x}{b - a} \sum_{k=1}^n f(c_k) && \Delta x = \frac{b - a}{n}, \text{ so } \frac{1}{n} = \frac{\Delta x}{b - a} \\ &= \frac{1}{b - a} \sum_{k=1}^n f(c_k) \Delta x \end{aligned}$$

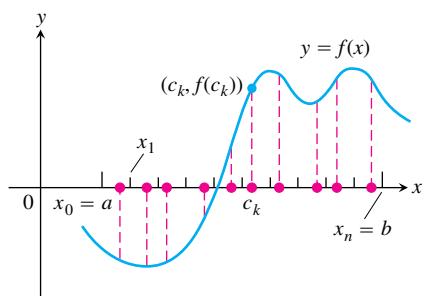


FIGURE 5.14 A sample of values of a function on an interval $[a, b]$.

The average is obtained by dividing a Riemann sum for f on $[a, b]$ by $(b - a)$. As we increase the size of the sample and let the norm of the partition approach zero, the average approaches $(1/(b - a)) \int_a^b f(x) dx$. Both points of view lead us to the following definition.

DEFINITION **The Average or Mean Value of a Function**

If f is integrable on $[a, b]$, then its **average value on $[a, b]$** , also called its **mean value**, is

$$\text{av}(f) = \frac{1}{b - a} \int_a^b f(x) dx.$$

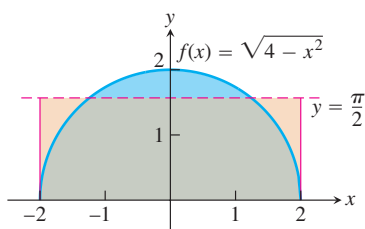


FIGURE 5.15 The average value of $f(x) = \sqrt{4 - x^2}$ on $[-2, 2]$ is $\pi/2$ (Example 5).

EXAMPLE 5 **Finding an Average Value**

Find the average value of $f(x) = \sqrt{4 - x^2}$ on $[-2, 2]$.

Solution We recognize $f(x) = \sqrt{4 - x^2}$ as a function whose graph is the upper semicircle of radius 2 centered at the origin (Figure 5.15).

The area between the semicircle and the x -axis from -2 to 2 can be computed using the geometry formula

$$\text{Area} = \frac{1}{2} \cdot \pi r^2 = \frac{1}{2} \cdot \pi (2)^2 = 2\pi.$$

Because f is nonnegative, the area is also the value of the integral of f from -2 to 2 ,

$$\int_{-2}^2 \sqrt{4 - x^2} dx = 2\pi.$$

Therefore, the average value of f is

$$\text{av}(f) = \frac{1}{2 - (-2)} \int_{-2}^2 \sqrt{4 - x^2} dx = \frac{1}{4} (2\pi) = \frac{\pi}{2}.$$

■

EXERCISES 5.3

Expressing Limits as Integrals

Express the limits in Exercises 1–8 as definite integrals.

- $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n c_k^2 \Delta x_k$, where P is a partition of $[0, 2]$
- $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n 2c_k^3 \Delta x_k$, where P is a partition of $[-1, 0]$
- $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (c_k^2 - 3c_k) \Delta x_k$, where P is a partition of $[-7, 5]$
- $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \left(\frac{1}{c_k}\right) \Delta x_k$, where P is a partition of $[1, 4]$
- $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \frac{1}{1 - c_k} \Delta x_k$, where P is a partition of $[2, 3]$
- $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n \sqrt{4 - c_k^2} \Delta x_k$, where P is a partition of $[0, 1]$
- $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (\sec c_k) \Delta x_k$, where P is a partition of $[-\pi/4, 0]$
- $\lim_{\|P\| \rightarrow 0} \sum_{k=1}^n (\tan c_k) \Delta x_k$, where P is a partition of $[0, \pi/4]$

Using Properties and Known Values to Find Other Integrals

9. Suppose that f and g are integrable and that

$$\int_1^2 f(x) dx = -4, \int_1^5 f(x) dx = 6, \int_1^5 g(x) dx = 8.$$

Use the rules in Table 5.3 to find

- | | |
|--------------------------------|---------------------------------|
| a. $\int_2^2 g(x) dx$ | b. $\int_5^1 g(x) dx$ |
| c. $\int_1^2 3f(x) dx$ | d. $\int_2^5 f(x) dx$ |
| e. $\int_1^5 [f(x) - g(x)] dx$ | f. $\int_1^5 [4f(x) - g(x)] dx$ |

10. Suppose that f and h are integrable and that

$$\int_1^9 f(x) dx = -1, \int_7^9 f(x) dx = 5, \int_7^9 h(x) dx = 4.$$

Use the rules in Table 5.3 to find

- | | |
|----------------------------------|--------------------------------|
| a. $\int_1^9 -2f(x) dx$ | b. $\int_7^9 [f(x) + h(x)] dx$ |
| c. $\int_7^9 [2f(x) - 3h(x)] dx$ | d. $\int_7^1 f(x) dx$ |
| e. $\int_1^7 f(x) dx$ | f. $\int_9^7 [h(x) - f(x)] dx$ |

11. Suppose that $\int_1^2 f(x) dx = 5$. Find

- | | |
|-----------------------|-------------------------------|
| a. $\int_1^2 f(u) du$ | b. $\int_1^2 \sqrt{3}f(z) dz$ |
| c. $\int_2^1 f(t) dt$ | d. $\int_1^2 [-f(x)] dx$ |

12. Suppose that $\int_{-3}^0 g(t) dt = \sqrt{2}$. Find

- | | |
|-----------------------------|---|
| a. $\int_0^{-3} g(t) dt$ | b. $\int_{-3}^0 g(u) du$ |
| c. $\int_{-3}^0 [-g(x)] dx$ | d. $\int_{-3}^0 \frac{g(r)}{\sqrt{2}} dr$ |

13. Suppose that f is integrable and that $\int_0^3 f(z) dz = 3$ and $\int_0^4 f(z) dz = 7$. Find

- | | |
|-----------------------|-----------------------|
| a. $\int_3^4 f(z) dz$ | b. $\int_4^3 f(t) dt$ |
|-----------------------|-----------------------|

14. Suppose that h is integrable and that $\int_{-1}^1 h(r) dr = 0$ and $\int_{-1}^3 h(r) dr = 6$. Find

- | | |
|-----------------------|------------------------|
| a. $\int_1^3 h(r) dr$ | b. $-\int_3^1 h(u) du$ |
|-----------------------|------------------------|

Using Area to Evaluate Definite Integrals

In Exercises 15–22, graph the integrands and use areas to evaluate the integrals.

- | | |
|---|-------------------------------------|
| 15. $\int_{-2}^4 \left(\frac{x}{2} + 3\right) dx$ | 16. $\int_{1/2}^{3/2} (-2x + 4) dx$ |
|---|-------------------------------------|

17. $\int_{-3}^3 \sqrt{9 - x^2} dx$

19. $\int_{-2}^1 |x| dx$

21. $\int_{-1}^1 (2 - |x|) dx$

18. $\int_{-4}^0 \sqrt{16 - x^2} dx$

20. $\int_{-1}^1 (1 - |x|) dx$

22. $\int_{-1}^1 (1 + \sqrt{1 - x^2}) dx$

Use areas to evaluate the integrals in Exercises 23–26.

23. $\int_0^b \frac{x}{2} dx, \quad b > 0$

24. $\int_0^b 4x dx, \quad b > 0$

25. $\int_a^b 2s ds, \quad 0 < a < b$

26. $\int_a^b 3t dt, \quad 0 < a < b$

Evaluations

Use the results of Equations (1) and (3) to evaluate the integrals in Exercises 27–38.

27. $\int_1^{\sqrt{2}} x dx$

28. $\int_{0.5}^{2.5} x dx$

29. $\int_{\pi}^{2\pi} \theta d\theta$

30. $\int_{\sqrt{2}}^{5\sqrt{2}} r dr$

31. $\int_0^{\sqrt[3]{7}} x^2 dx$

32. $\int_0^{0.3} s^2 ds$

33. $\int_0^{1/2} t^2 dt$

34. $\int_0^{\pi/2} \theta^2 d\theta$

35. $\int_a^{2a} x dx$

36. $\int_a^{\sqrt{3}a} x dx$

37. $\int_0^{\sqrt[3]{b}} x^2 dx$

38. $\int_0^{3b} x^2 dx$

Use the rules in Table 5.3 and Equations (1)–(3) to evaluate the integrals in Exercises 39–50.

39. $\int_3^1 7 dx$

40. $\int_0^{-2} \sqrt{2} dx$

41. $\int_0^2 5x dx$

42. $\int_3^5 \frac{x}{8} dx$

43. $\int_0^2 (2t - 3) dt$

44. $\int_0^{\sqrt{2}} (t - \sqrt{2}) dt$

45. $\int_2^1 \left(1 + \frac{z}{2}\right) dz$

46. $\int_3^0 (2z - 3) dz$

47. $\int_1^2 3u^2 du$

48. $\int_{1/2}^1 24u^2 du$

49. $\int_0^2 (3x^2 + x - 5) dx$

50. $\int_1^0 (3x^2 + x - 5) dx$

Finding Area

In Exercises 51–54 use a definite integral to find the area of the region between the given curve and the x -axis on the interval $[0, b]$.

51. $y = 3x^2$

52. $y = \pi x^2$

53. $y = 2x$

54. $y = \frac{x}{2} + 1$

Average Value

In Exercises 55–62, graph the function and find its average value over the given interval.

55. $f(x) = x^2 - 1$ on $[0, \sqrt{3}]$

56. $f(x) = -\frac{x^2}{2}$ on $[0, 3]$ 57. $f(x) = -3x^2 - 1$ on $[0, 1]$

58. $f(x) = 3x^2 - 3$ on $[0, 1]$

59. $f(t) = (t - 1)^2$ on $[0, 3]$

60. $f(t) = t^2 - t$ on $[-2, 1]$

61. $g(x) = |x| - 1$ on a. $[-1, 1]$, b. $[1, 3]$, and c. $[-1, 3]$

62. $h(x) = -|x|$ on a. $[-1, 0]$, b. $[0, 1]$, and c. $[-1, 1]$

Theory and Examples

63. What values of a and b maximize the value of

$$\int_a^b (x - x^2) dx$$

(Hint: Where is the integrand positive?)

64. What values of a and b minimize the value of

$$\int_a^b (x^4 - 2x^2) dx$$

65. Use the Max-Min Inequality to find upper and lower bounds for the value of

$$\int_0^1 \frac{1}{1 + x^2} dx.$$

66. (Continuation of Exercise 65) Use the Max-Min Inequality to find upper and lower bounds for

$$\int_0^{0.5} \frac{1}{1 + x^2} dx \quad \text{and} \quad \int_{0.5}^1 \frac{1}{1 + x^2} dx.$$

Add these to arrive at an improved estimate of

$$\int_0^1 \frac{1}{1 + x^2} dx.$$

67. Show that the value of $\int_0^1 \sin(x^2) dx$ cannot possibly be 2.

68. Show that the value of $\int_1^0 \sqrt{x + 8} dx$ lies between $2\sqrt{2} \approx 2.8$ and 3.

69. **Integrals of nonnegative functions** Use the Max-Min Inequality to show that if f is integrable then

$$f(x) \geq 0 \quad \text{on} \quad [a, b] \quad \Rightarrow \quad \int_a^b f(x) dx \geq 0.$$

70. **Integrals of nonpositive functions** Show that if f is integrable then

$$f(x) \leq 0 \quad \text{on} \quad [a, b] \quad \Rightarrow \quad \int_a^b f(x) dx \leq 0.$$

71. Use the inequality $\sin x \leq x$, which holds for $x \geq 0$, to find an upper bound for the value of $\int_0^1 \sin x dx$.

72. The inequality $\sec x \geq 1 + (x^2/2)$ holds on $(-\pi/2, \pi/2)$. Use it to find a lower bound for the value of $\int_0^1 \sec x dx$.

73. If $\text{av}(f)$ really is a typical value of the integrable function $f(x)$ on $[a, b]$, then the number $\text{av}(f)$ should have the same integral over $[a, b]$ that f does. Does it? That is, does

$$\int_a^b \text{av}(f) dx = \int_a^b f(x) dx?$$

Give reasons for your answer.

74. It would be nice if average values of integrable functions obeyed the following rules on an interval $[a, b]$.

a. $\text{av}(f + g) = \text{av}(f) + \text{av}(g)$

b. $\text{av}(kf) = k \text{av}(f)$ (any number k)

c. $\text{av}(f) \leq \text{av}(g)$ if $f(x) \leq g(x)$ on $[a, b]$.

Do these rules ever hold? Give reasons for your answers.

75. Use limits of Riemann sums as in Example 4a to establish Equation (2).

76. Use limits of Riemann sums as in Example 4a to establish Equation (3).

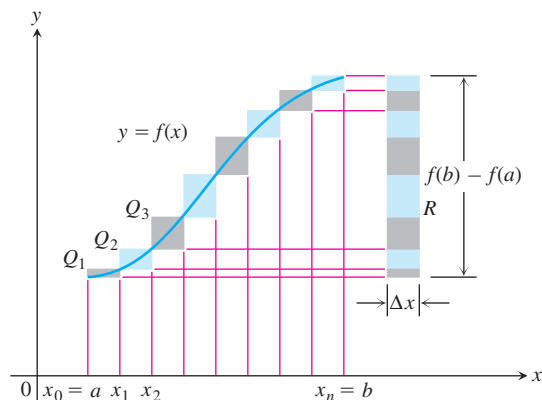
77. Upper and lower sums for increasing functions

a. Suppose the graph of a continuous function $f(x)$ rises steadily as x moves from left to right across an interval $[a, b]$. Let P be a partition of $[a, b]$ into n subintervals of length $\Delta x = (b - a)/n$. Show by referring to the accompanying figure that the difference between the upper and lower sums for f on this partition can be represented graphically as the area of a rectangle R whose dimensions are $[f(b) - f(a)]$ by Δx . (Hint: The difference $U - L$ is the sum of areas of rectangles whose diagonals $Q_0Q_1, Q_1Q_2, \dots, Q_{n-1}Q_n$ lie along the curve. There is no overlapping when these rectangles are shifted horizontally onto R .)

b. Suppose that instead of being equal, the lengths Δx_k of the subintervals of the partition of $[a, b]$ vary in size. Show that

$$U - L \leq |f(b) - f(a)| \Delta x_{\max},$$

where $\Delta x_{\max}$ is the norm of P , and hence that $\lim_{\|P\| \rightarrow 0} (U - L) = 0$.



78. Upper and lower sums for decreasing functions (Continuation of Exercise 77)

- a. Draw a figure like the one in Exercise 77 for a continuous function $f(x)$ whose values decrease steadily as x moves from left to right across the interval $[a, b]$. Let P be a partition of $[a, b]$ into subintervals of equal length. Find an expression for $U - L$ that is analogous to the one you found for $U - L$ in Exercise 77a.
- b. Suppose that instead of being equal, the lengths Δx_k of the subintervals of P vary in size. Show that the inequality

$$U - L \leq |f(b) - f(a)| \Delta x_{\max}$$

of Exercise 77b still holds and hence that $\lim_{\|P\| \rightarrow 0} (U - L) = 0$.

79. Use the formula

$$\begin{aligned} \sin h + \sin 2h + \sin 3h + \cdots + \sin mh \\ = \frac{\cos(h/2) - \cos((m + (1/2))h)}{2 \sin(h/2)} \end{aligned}$$

to find the area under the curve $y = \sin x$ from $x = 0$ to $x = \pi/2$ in two steps:

- a. Partition the interval $[0, \pi/2]$ into n subintervals of equal length and calculate the corresponding upper sum U ; then
- b. Find the limit of U as $n \rightarrow \infty$ and $\Delta x = (b - a)/n \rightarrow 0$.
- 80.** Suppose that f is continuous and nonnegative over $[a, b]$, as in the figure at the right. By inserting points

$$x_1, x_2, \dots, x_{k-1}, x_k, \dots, x_{n-1}$$

as shown, divide $[a, b]$ into n subintervals of lengths $\Delta x_1 = x_1 - a$, $\Delta x_2 = x_2 - x_1, \dots, \Delta x_n = b - x_{n-1}$, which need not be equal.

- a. If $m_k = \min \{f(x) \text{ for } x \text{ in the } k\text{th subinterval}\}$, explain the connection between the *lower sum*

$$L = m_1 \Delta x_1 + m_2 \Delta x_2 + \cdots + m_n \Delta x_n$$

and the shaded region in the first part of the figure.

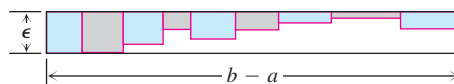
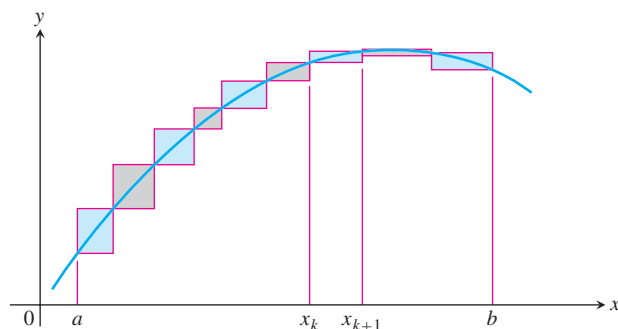
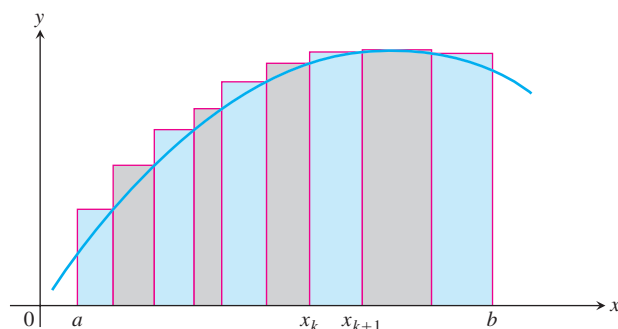
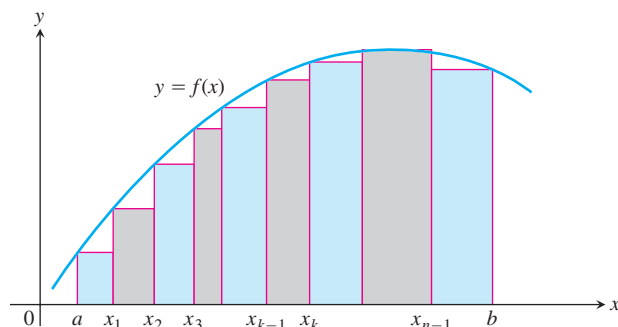
- b. If $M_k = \max \{f(x) \text{ for } x \text{ in the } k\text{th subinterval}\}$, explain the connection between the *upper sum*

$$U = M_1 \Delta x_1 + M_2 \Delta x_2 + \cdots + M_n \Delta x_n$$

and the shaded region in the second part of the figure.

- c. Explain the connection between $U - L$ and the shaded regions along the curve in the third part of the figure.

- 81.** We say f is **uniformly continuous** on $[a, b]$ if given any $\epsilon > 0$ there is a $\delta > 0$ such that if x_1, x_2 are in $[a, b]$ and $|x_1 - x_2| < \delta$ then $|f(x_1) - f(x_2)| < \epsilon$. It can be shown that a continuous function on $[a, b]$ is uniformly continuous. Use this and the figure at the right to show that if f is continuous and $\epsilon > 0$ is given, it is possible to make $U - L \leq \epsilon \cdot (b - a)$ by making the largest of the Δx_k 's sufficiently small.
- 82.** If you average 30 mi/h on a 150-mi trip and then return over the same 150 mi at the rate of 50 mi/h, what is your average speed for the trip? Give reasons for your answer. (Source: David H.



Pleacher, *The Mathematics Teacher*, Vol. 85, No. 6, pp. 445–446, September 1992.)

COMPUTER EXPLORATIONS

Finding Riemann Sums

If your CAS can draw rectangles associated with Riemann sums, use it to draw rectangles associated with Riemann sums that converge to the integrals in Exercises 83–88. Use $n = 4, 10, 20$, and 50 subintervals of equal length in each case.

83. $\int_0^1 (1 - x) dx = \frac{1}{2}$

84. $\int_0^1 (x^2 + 1) dx = \frac{4}{3}$

$$85. \int_{-\pi}^{\pi} \cos x \, dx = 0$$

$$86. \int_0^{\pi/4} \sec^2 x \, dx = 1$$

$$87. \int_{-1}^1 |x| \, dx = 1$$

$$88. \int_1^2 \frac{1}{x} \, dx \text{ (The integral's value is about 0.693.)}$$

Average Value

In Exercises 89–92, use a CAS to perform the following steps:

- Plot the functions over the given interval.
- Partition the interval into $n = 100, 200$, and 1000 subintervals of equal length, and evaluate the function at the midpoint of each subinterval.

c. Compute the average value of the function values generated in part (b).

d. Solve the equation $f(x) = (\text{average value})$ for x using the average value calculated in part (c) for the $n = 1000$ partitioning.

$$89. f(x) = \sin x \quad \text{on} \quad [0, \pi]$$

$$90. f(x) = \sin^2 x \quad \text{on} \quad [0, \pi]$$

$$91. f(x) = x \sin \frac{1}{x} \quad \text{on} \quad \left[\frac{\pi}{4}, \pi \right]$$

$$92. f(x) = x \sin^2 \frac{1}{x} \quad \text{on} \quad \left[\frac{\pi}{4}, \pi \right]$$

5.4 The Fundamental Theorem of Calculus

HISTORICAL BIOGRAPHY

Sir Isaac Newton
(1642–1727)

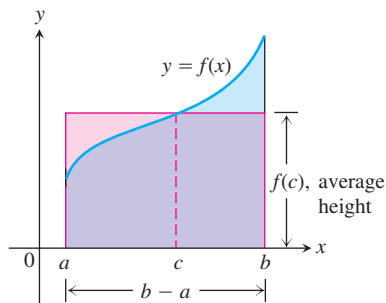


FIGURE 5.16 The value $f(c)$ in the Mean Value Theorem is, in a sense, the average (or *mean*) height of f on $[a, b]$. When $f \geq 0$, the area of the rectangle is the area under the graph of f from a to b ,

$$f(c)(b - a) = \int_a^b f(x) \, dx.$$

In this section we present the Fundamental Theorem of Calculus, which is the central theorem of integral calculus. It connects integration and differentiation, enabling us to compute integrals using an antiderivative of the integrand function rather than by taking limits of Riemann sums as we did in Section 5.3. Leibniz and Newton exploited this relationship and started mathematical developments that fueled the scientific revolution for the next 200 years.

Along the way, we present the integral version of the Mean Value Theorem, which is another important theorem of integral calculus and used to prove the Fundamental Theorem.

Mean Value Theorem for Definite Integrals

In the previous section, we defined the average value of a continuous function over a closed interval $[a, b]$ as the definite integral $\int_a^b f(x) \, dx$ divided by the length or width $b - a$ of the interval. The Mean Value Theorem for Definite Integrals asserts that this average value is *always* taken on at least once by the function f in the interval.

The graph in Figure 5.16 shows a *positive* continuous function $y = f(x)$ defined over the interval $[a, b]$. Geometrically, the Mean Value Theorem says that there is a number c in $[a, b]$ such that the rectangle with height equal to the average value $f(c)$ of the function and base width $b - a$ has exactly the same area as the region beneath the graph of f from a to b .

THEOREM 3 The Mean Value Theorem for Definite Integrals

If f is continuous on $[a, b]$, then at some point c in $[a, b]$,

$$f(c) = \frac{1}{b - a} \int_a^b f(x) \, dx.$$

Proof If we divide both sides of the Max-Min Inequality (Table 5.3, Rule 6) by $(b - a)$, we obtain

$$\min f \leq \frac{1}{b-a} \int_a^b f(x) \, dx \leq \max f.$$

Since f is continuous, the Intermediate Value Theorem for Continuous Functions (Section 2.6) says that f must assume every value between $\min f$ and $\max f$. It must therefore assume the value $(1/(b-a)) \int_a^b f(x) \, dx$ at some point c in $[a, b]$. ■

The continuity of f is important here. It is possible that a discontinuous function never equals its average value (Figure 5.17).

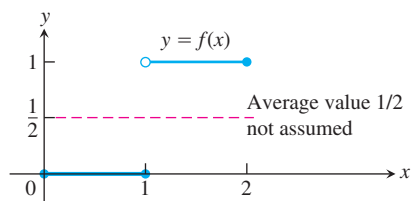


FIGURE 5.17 A discontinuous function need not assume its average value.

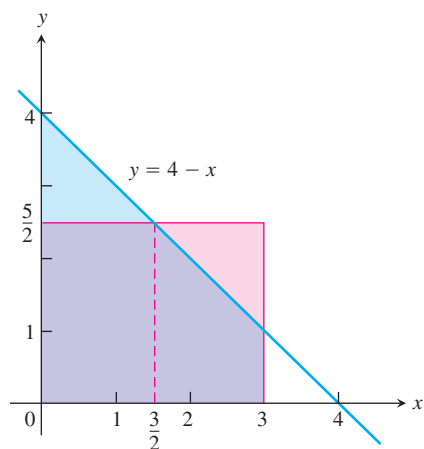


FIGURE 5.18 The area of the rectangle with base $[0, 3]$ and height $5/2$ (the average value of the function $f(x) = 4 - x$) is equal to the area between the graph of f and the x -axis from 0 to 3 (Example 1).

EXAMPLE 1 Applying the Mean Value Theorem for Integrals

Find the average value of $f(x) = 4 - x$ on $[0, 3]$ and where f actually takes on this value at some point in the given domain.

Solution

$$\begin{aligned} \text{av}(f) &= \frac{1}{b-a} \int_a^b f(x) \, dx \\ &= \frac{1}{3-0} \int_0^3 (4-x) \, dx = \frac{1}{3} \left(\int_0^3 4 \, dx - \int_0^3 x \, dx \right) \\ &= \frac{1}{3} \left(4(3-0) - \left(\frac{3^2}{2} - \frac{0^2}{2} \right) \right) \\ &= 4 - \frac{3}{2} = \frac{5}{2}. \end{aligned}$$

Section 5.3, Eqs. (1) and (2)

The average value of $f(x) = 4 - x$ over $[0, 3]$ is $5/2$. The function assumes this value when $4 - x = 5/2$ or $x = 3/2$. (Figure 5.18) ■

In Example 1, we actually found a point c where f assumed its average value by setting $f(x)$ equal to the calculated average value and solving for x . It's not always possible to solve easily for the value c . What else can we learn from the Mean Value Theorem for integrals? Here's an example.

EXAMPLE 2 Show that if f is continuous on $[a, b]$, $a \neq b$, and if

$$\int_a^b f(x) \, dx = 0,$$

then $f(x) = 0$ at least once in $[a, b]$.

Solution The average value of f on $[a, b]$ is

$$\text{av}(f) = \frac{1}{b-a} \int_a^b f(x) \, dx = \frac{1}{b-a} \cdot 0 = 0.$$

By the Mean Value Theorem, f assumes this value at some point $c \in [a, b]$. ■

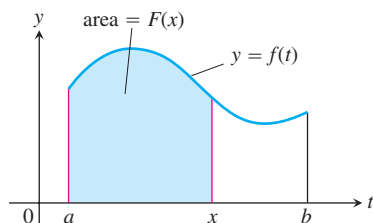


FIGURE 5.19 The function $F(x)$ defined by Equation (1) gives the area under the graph of f from a to x when f is nonnegative and $x > a$.

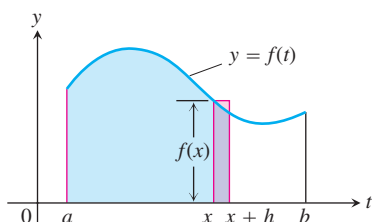


FIGURE 5.20 In Equation (1), $F(x)$ is the area to the left of x . Also, $F(x + h)$ is the area to the left of $x + h$. The difference quotient $[F(x + h) - F(x)]/h$ is then approximately equal to $f(x)$, the height of the rectangle shown here.

Fundamental Theorem, Part 1

If $f(t)$ is an integrable function over a finite interval I , then the integral from any fixed number $a \in I$ to another number $x \in I$ defines a new function F whose value at x is

$$F(x) = \int_a^x f(t) dt. \quad (1)$$

For example, if f is nonnegative and x lies to the right of a , then $F(x)$ is the area under the graph from a to x (Figure 5.19). The variable x is the upper limit of integration of an integral, but F is just like any other real-valued function of a real variable. For each value of the input x , there is a well-defined numerical output, in this case the definite integral of f from a to x .

Equation (1) gives a way to define new functions, but its importance now is the connection it makes between integrals and derivatives. If f is any continuous function, then the Fundamental Theorem asserts that F is a differentiable function of x whose derivative is f itself. At every value of x ,

$$\frac{d}{dx} F(x) = \frac{d}{dx} \int_a^x f(t) dt = f(x).$$

To gain some insight into why this result holds, we look at the geometry behind it.

If $f \geq 0$ on $[a, b]$, then the computation of $F'(x)$ from the definition of the derivative means taking the limit as $h \rightarrow 0$ of the difference quotient

$$\frac{F(x + h) - F(x)}{h}.$$

For $h > 0$, the numerator is obtained by subtracting two areas, so it is the area under the graph of f from x to $x + h$ (Figure 5.20). If h is small, this area is approximately equal to the area of the rectangle of height $f(x)$ and width h , which can be seen from Figure 5.20. That is,

$$F(x + h) - F(x) \approx hf(x).$$

Dividing both sides of this approximation by h and letting $h \rightarrow 0$, it is reasonable to expect that

$$F'(x) = \lim_{h \rightarrow 0} \frac{F(x + h) - F(x)}{h} = f(x).$$

This result is true even if the function f is not positive, and it forms the first part of the Fundamental Theorem of Calculus.

THEOREM 4 The Fundamental Theorem of Calculus Part 1

If f is continuous on $[a, b]$ then $F(x) = \int_a^x f(t) dt$ is continuous on $[a, b]$ and differentiable on (a, b) and its derivative is $f(x)$;

$$F'(x) = \frac{d}{dx} \int_a^x f(t) dt = f(x). \quad (2)$$